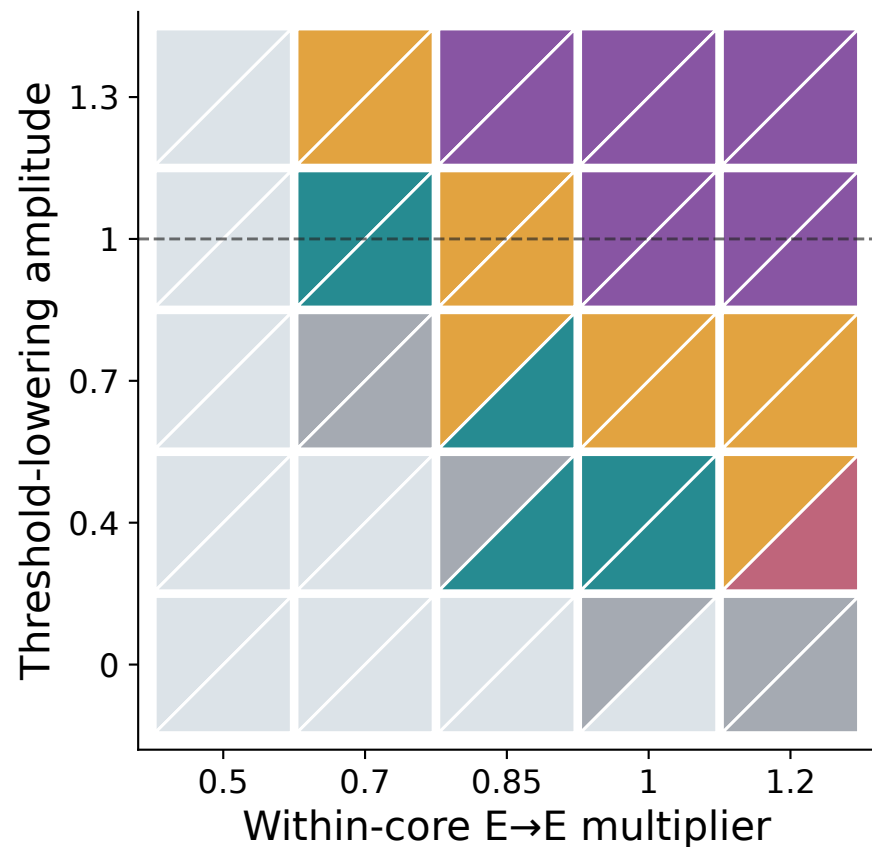


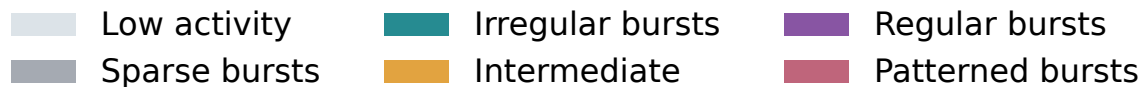
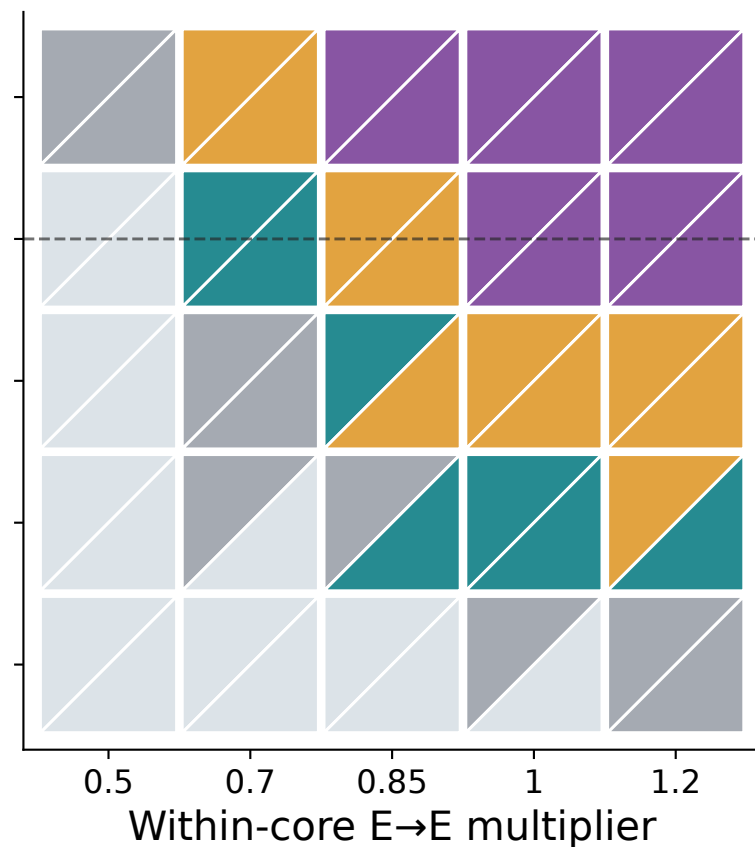
Native SNN activity regions

Each square: two noise seeds · dashed row: depth = 1 · finite-time labels

Core A



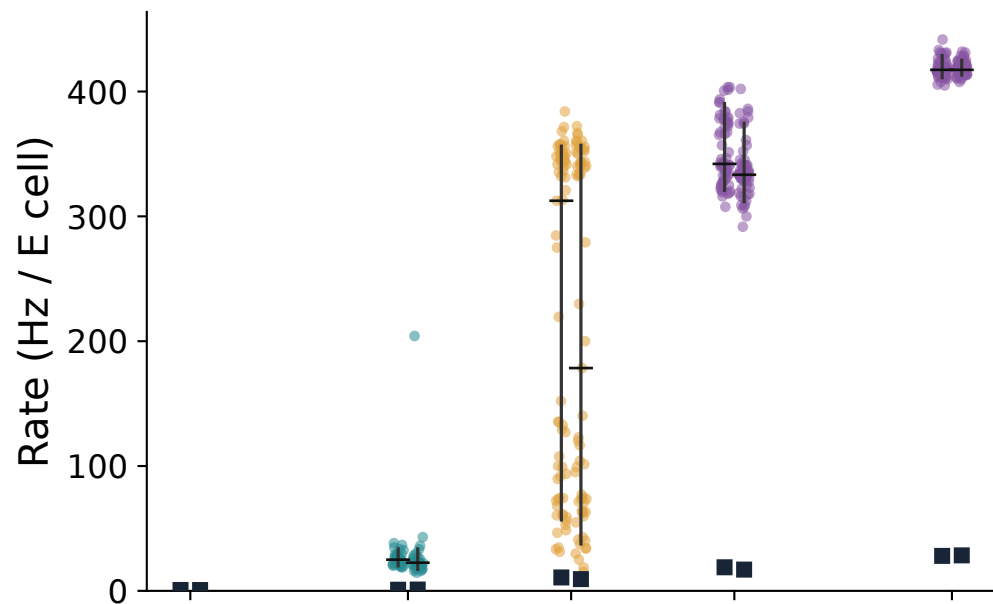
Core B



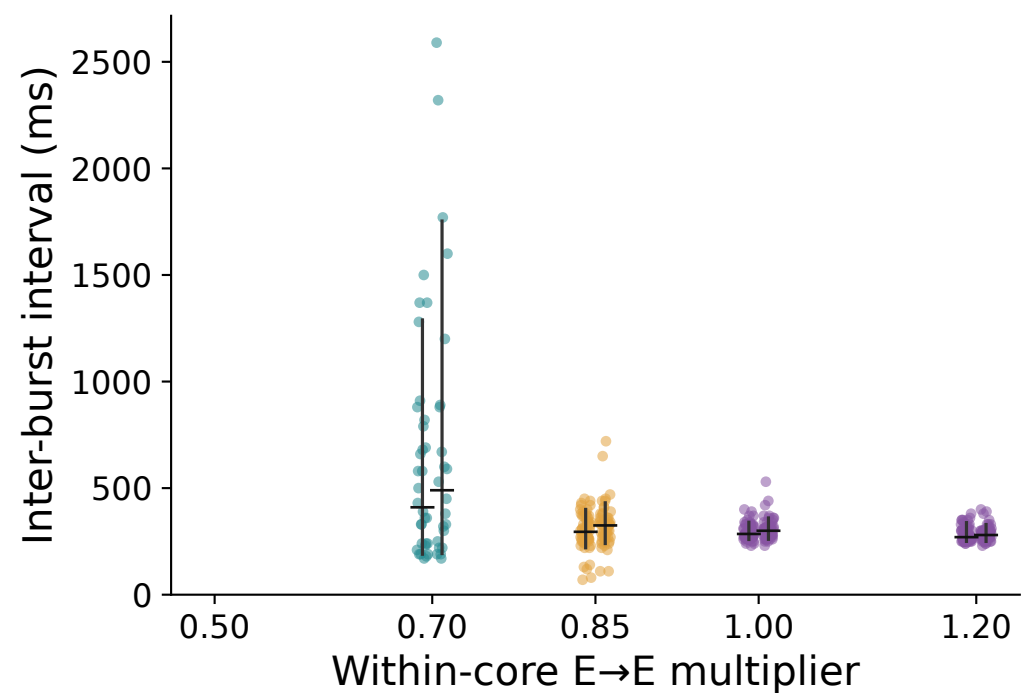
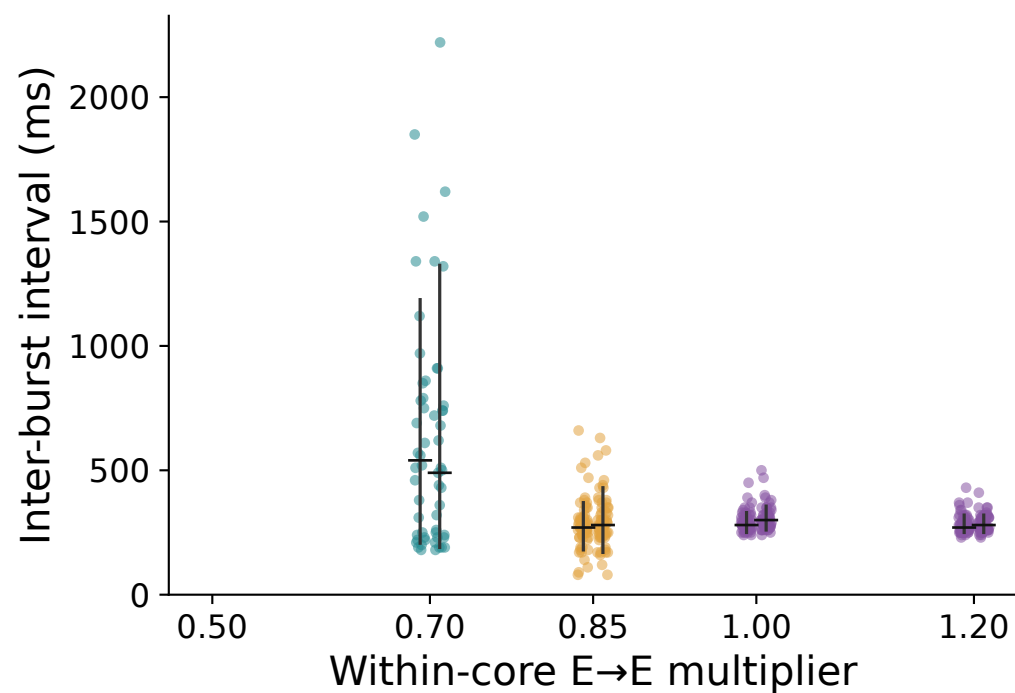
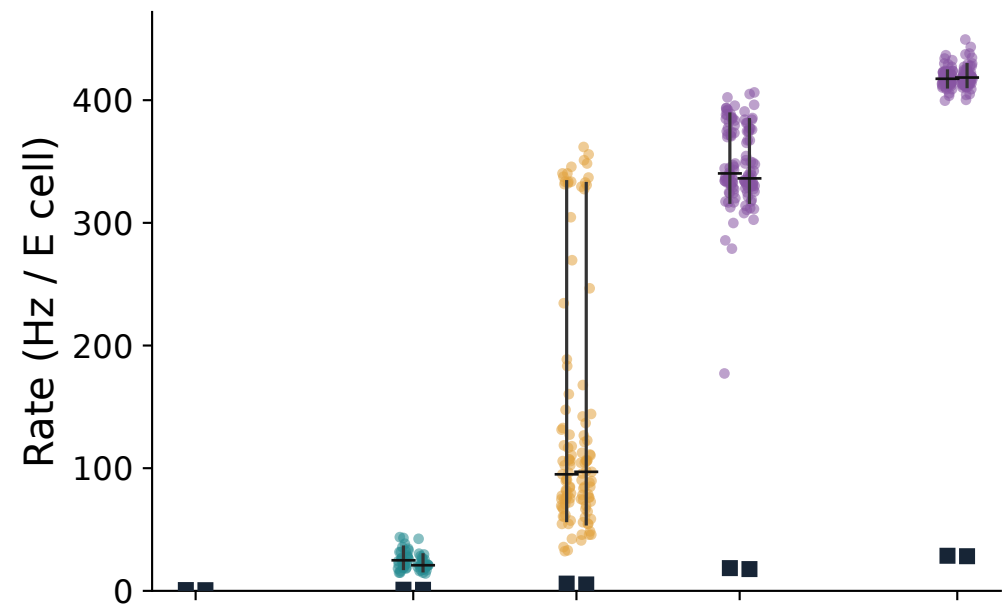
Burst onset and event-to-event variability

Dots: native event peaks / intervals · bars: within-run 10–90% · squares: mean firing rate

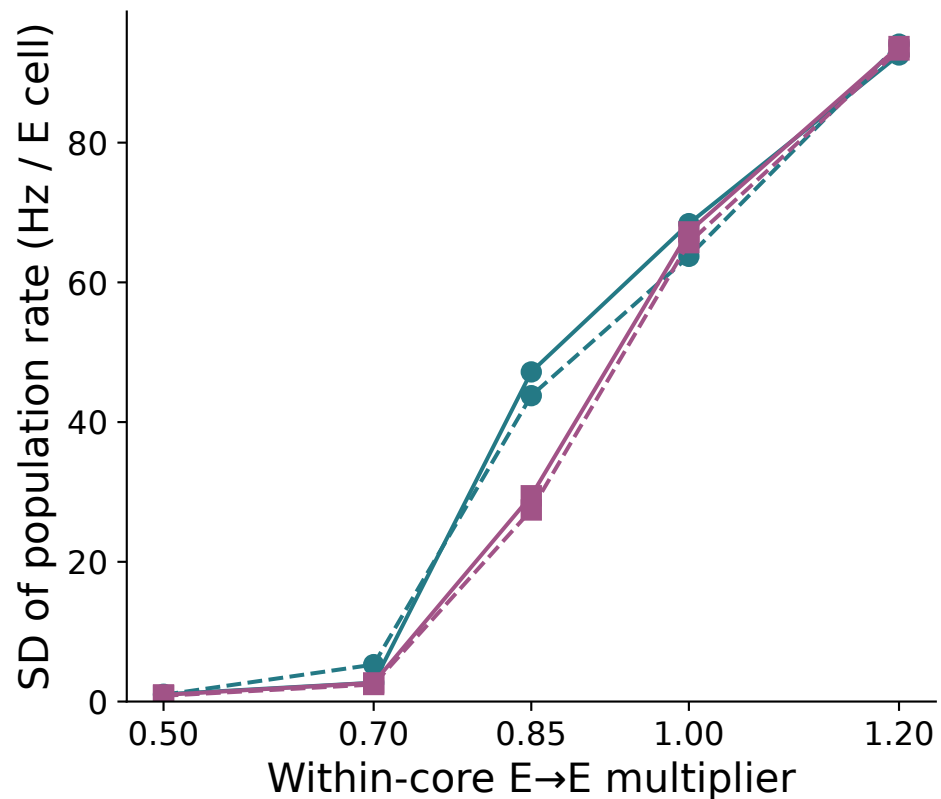
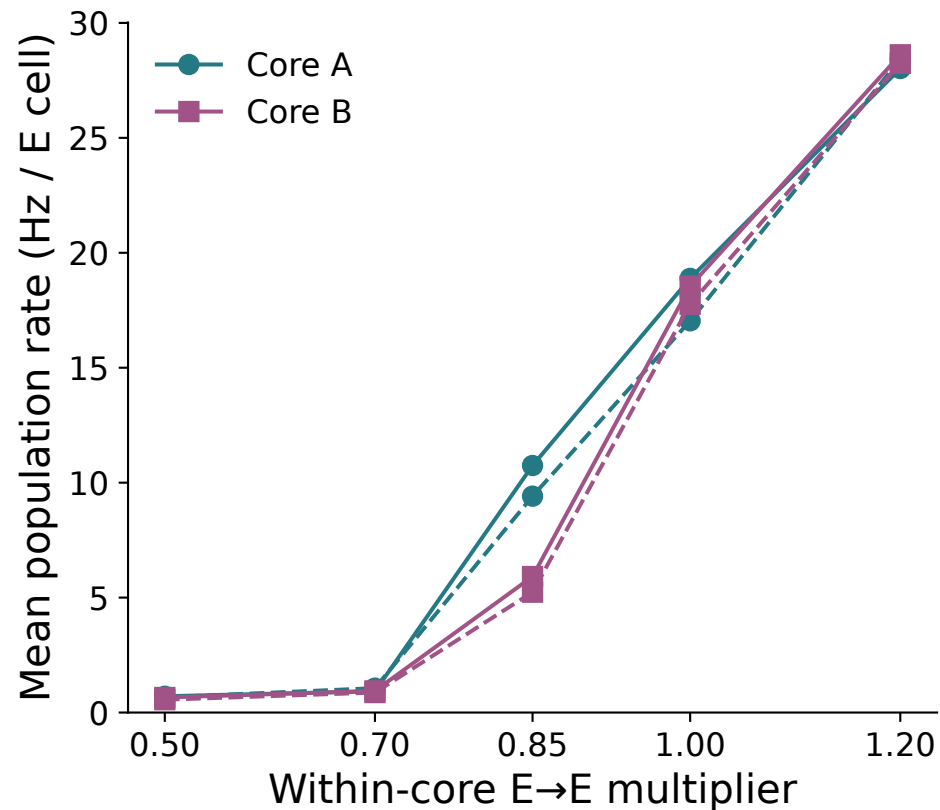
Core A



Core B



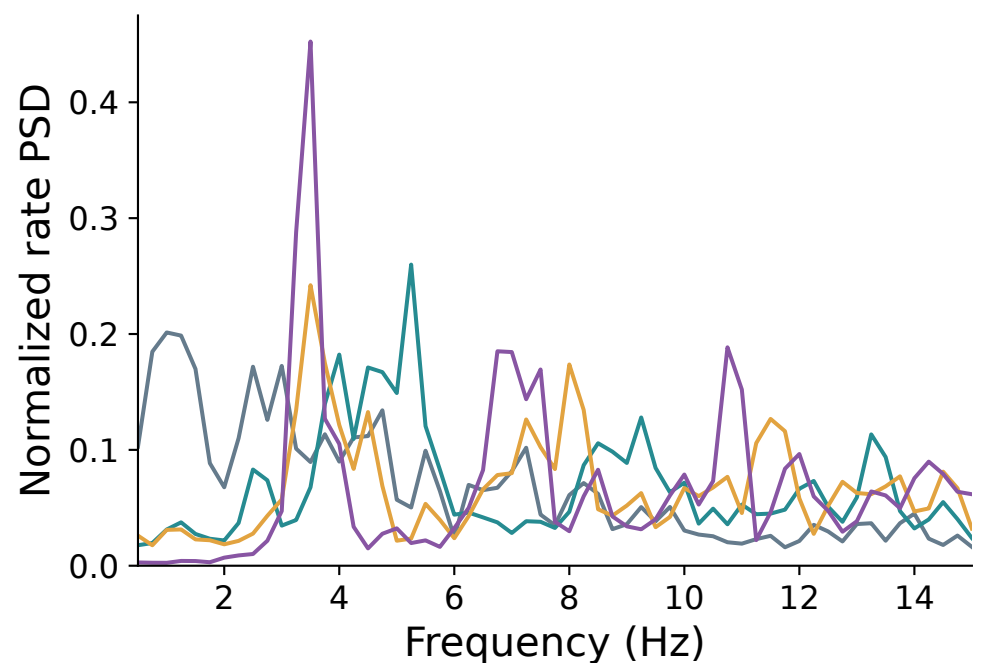
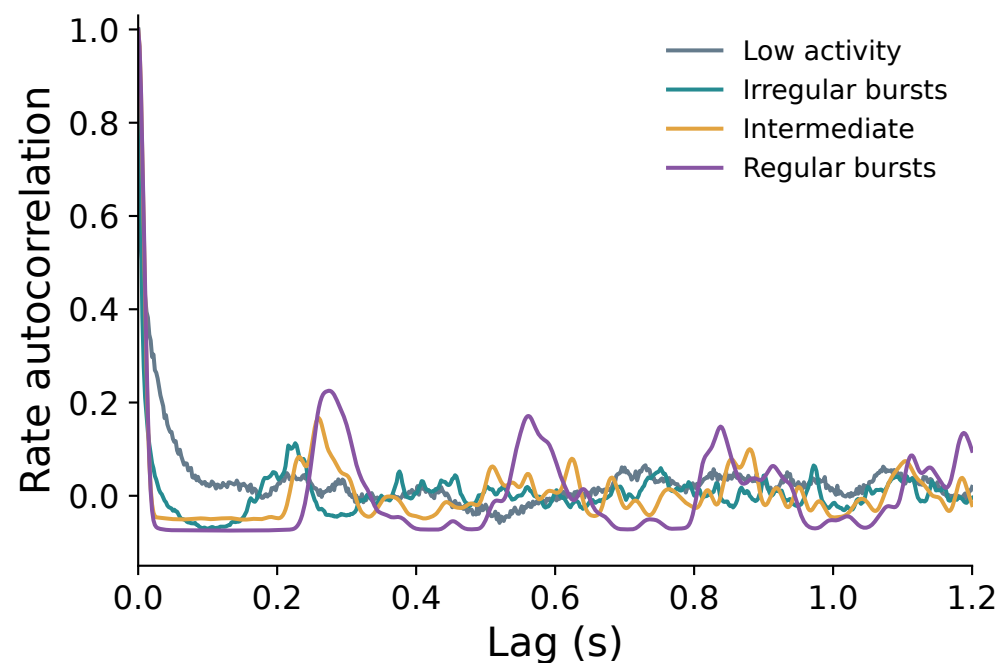
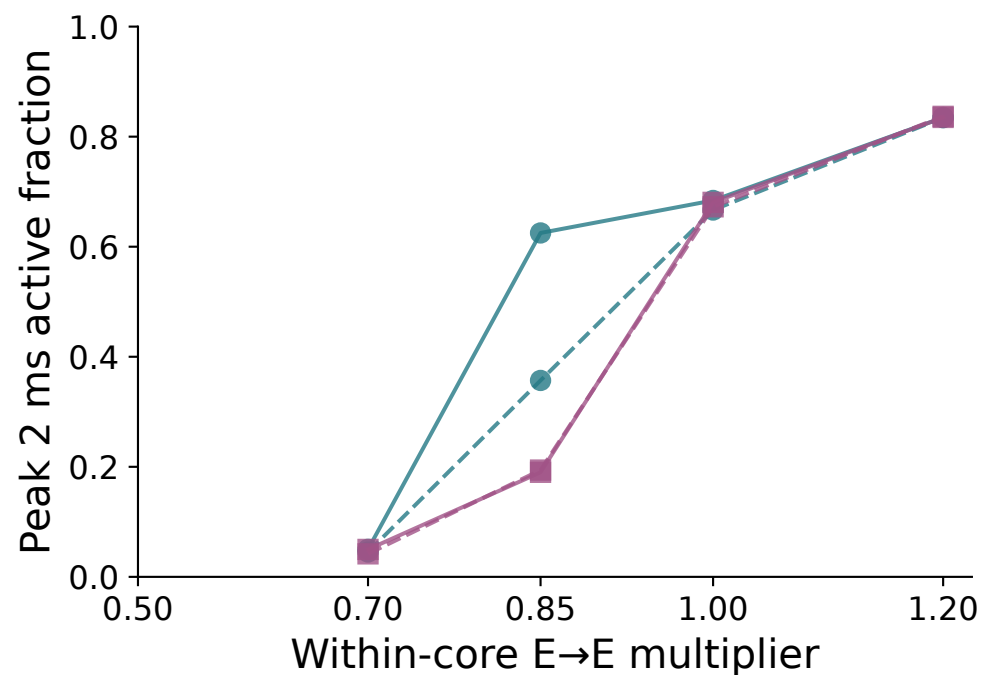
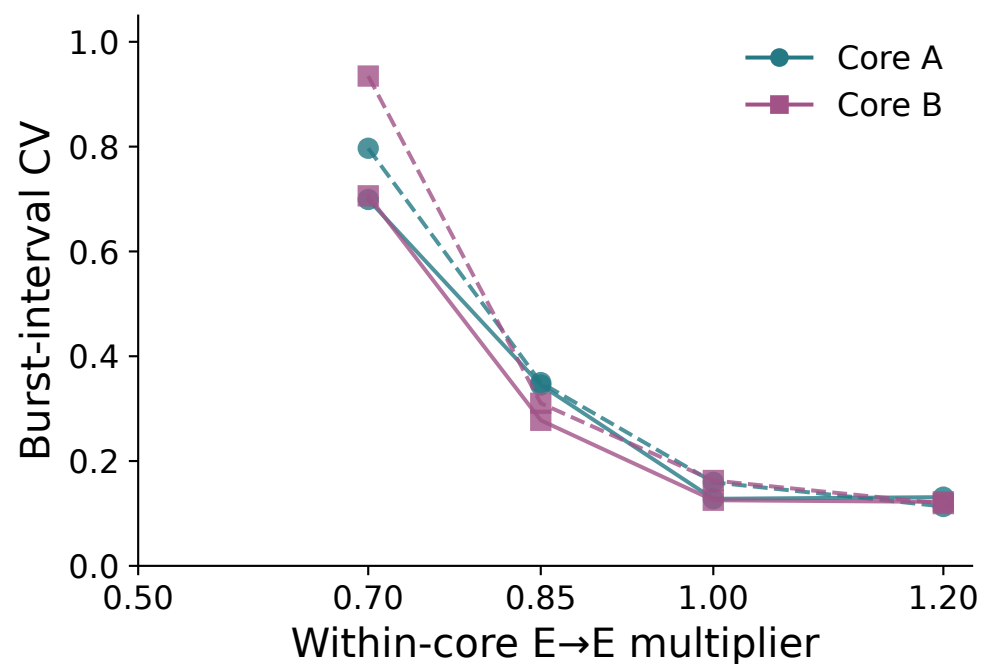
Population response without burst-detection thresholds



All native 2 ms bins, 2-20 s · depth = 1 · two cores × two noise seeds

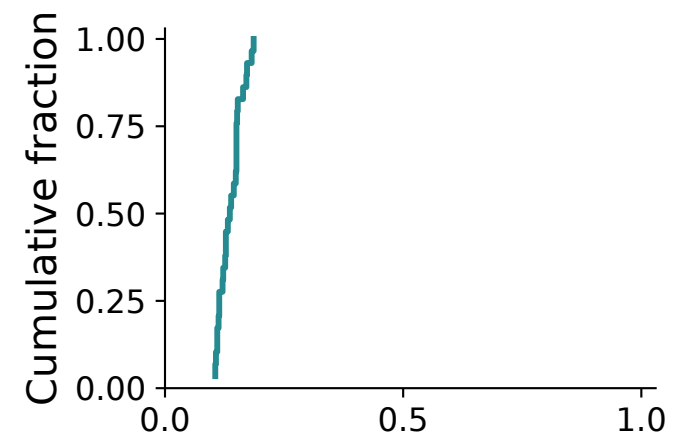
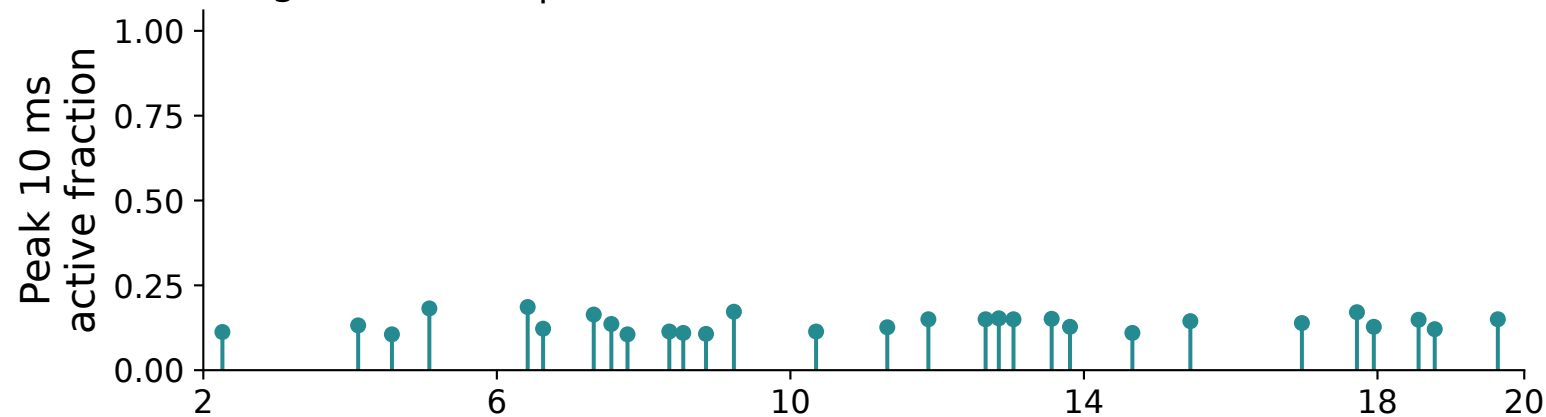
Timing, recruitment and population rhythmicity

Top: both cores $\times$ two noise seeds · bottom: fixed Core A, seed 848101

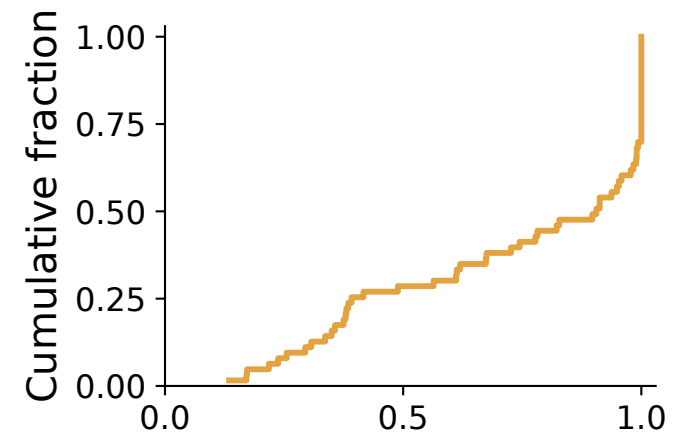
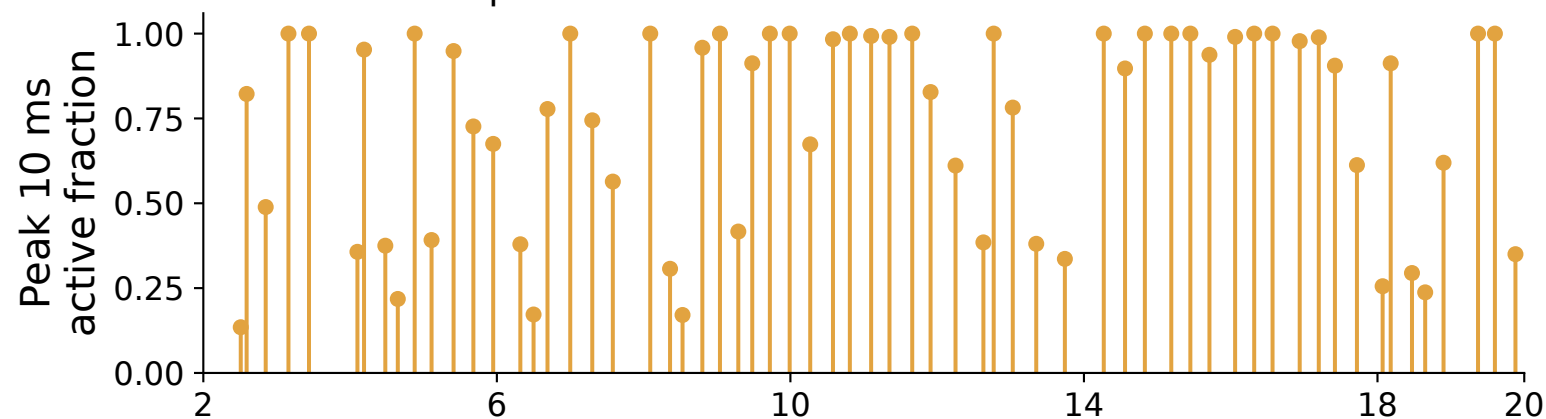


Intermediate bursts: variable recruitment

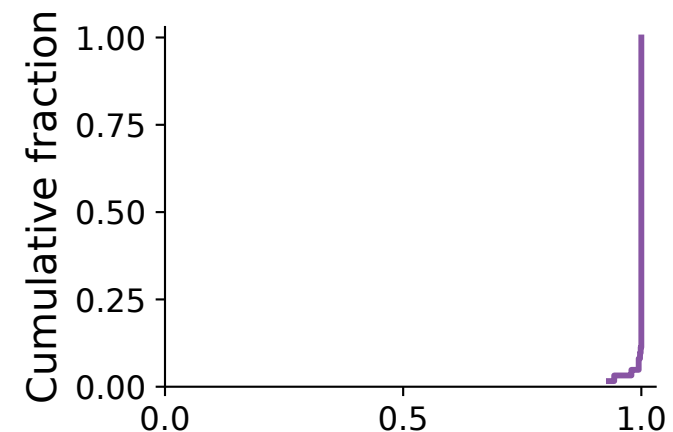
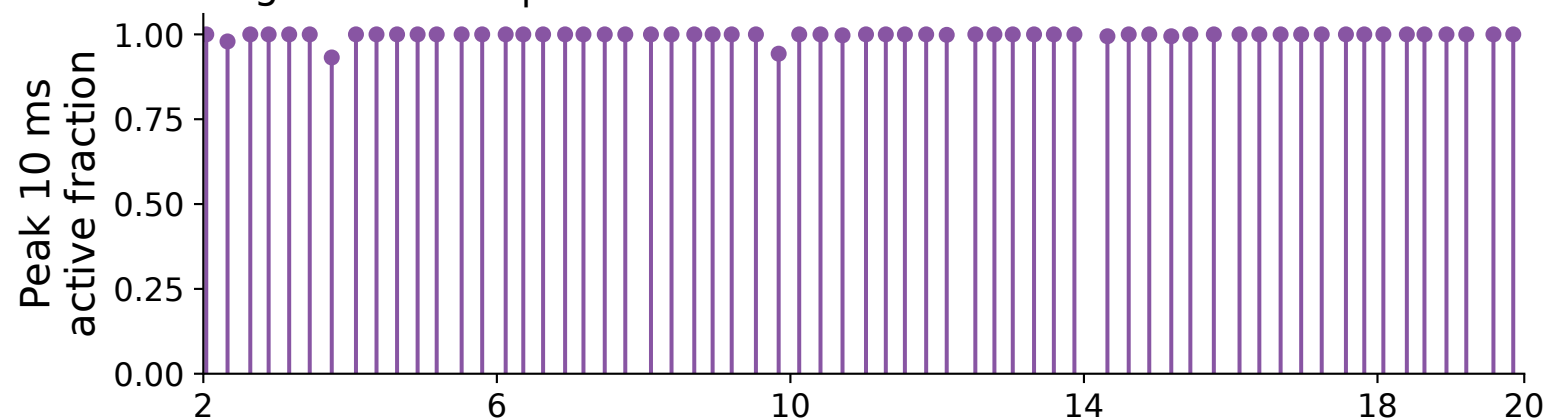
Irregular bursts | $E \rightarrow E = 0.7$



Intermediate | $E \rightarrow E = 0.85$



Regular bursts | $E \rightarrow E = 1$

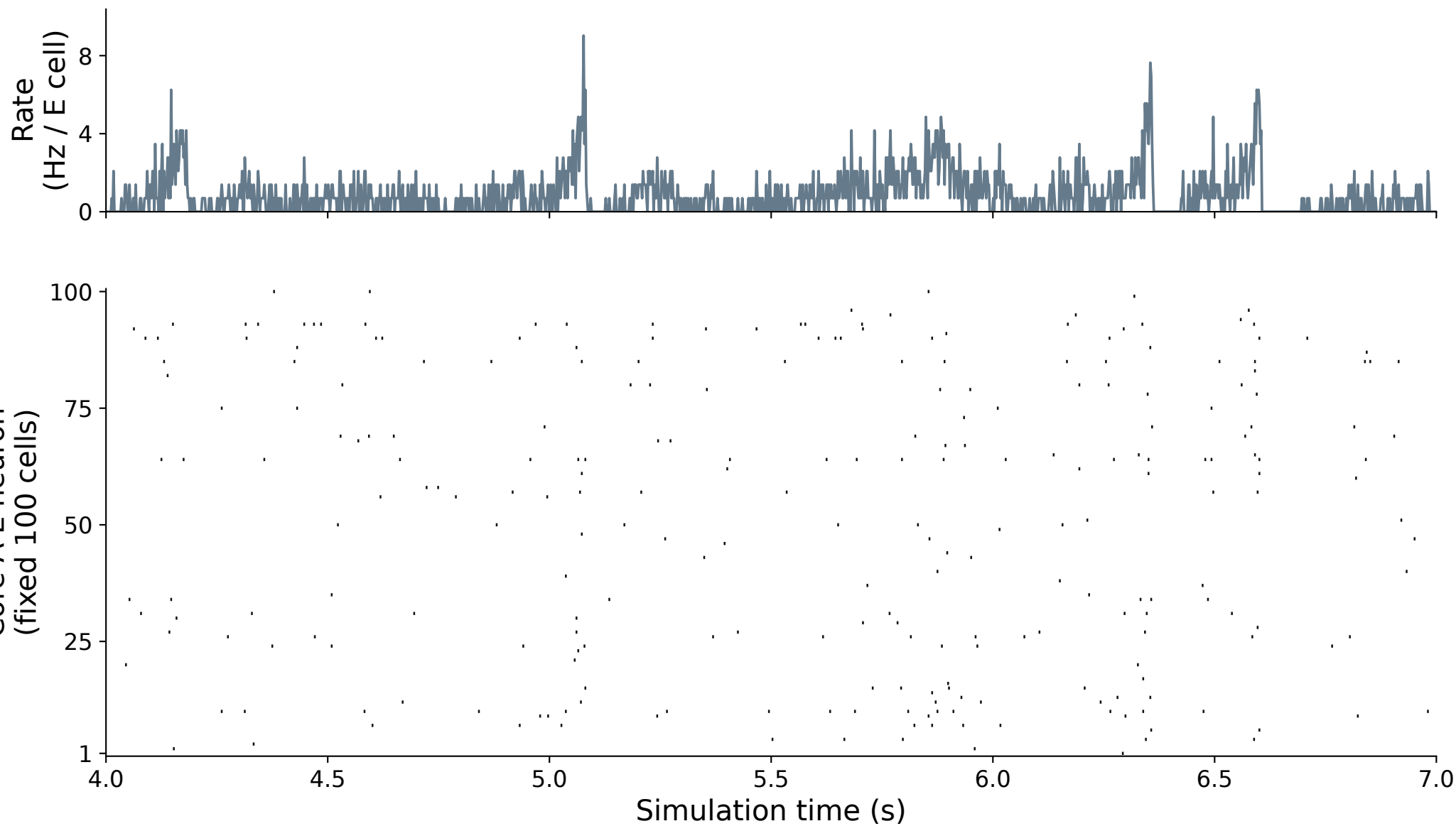


Simulation time (s)

Peak active fraction

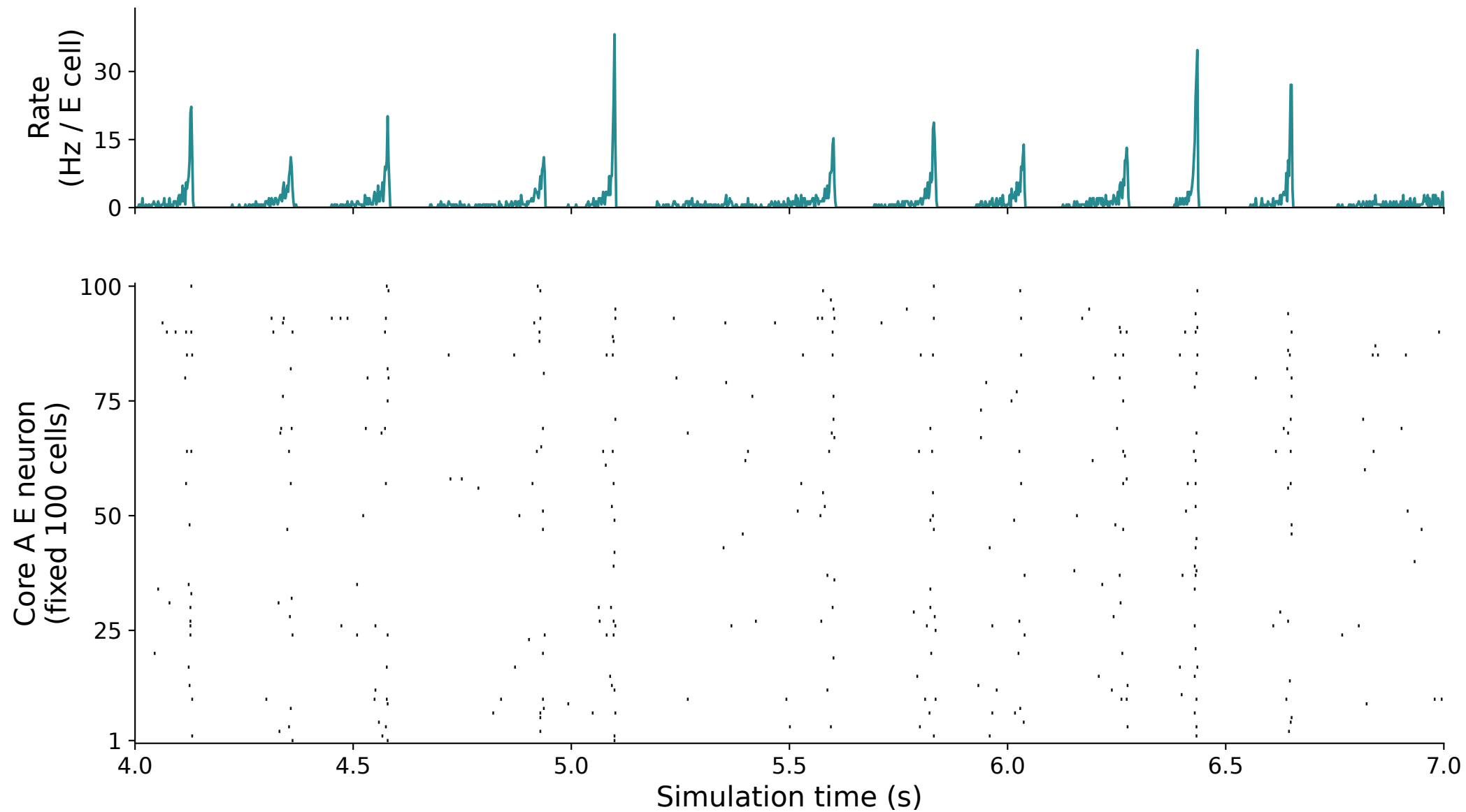
Low activity | $E \rightarrow E = 0.5$

Rate: all 720 Core A E cells · raster: occupied 2 ms bins · same time and cells across states



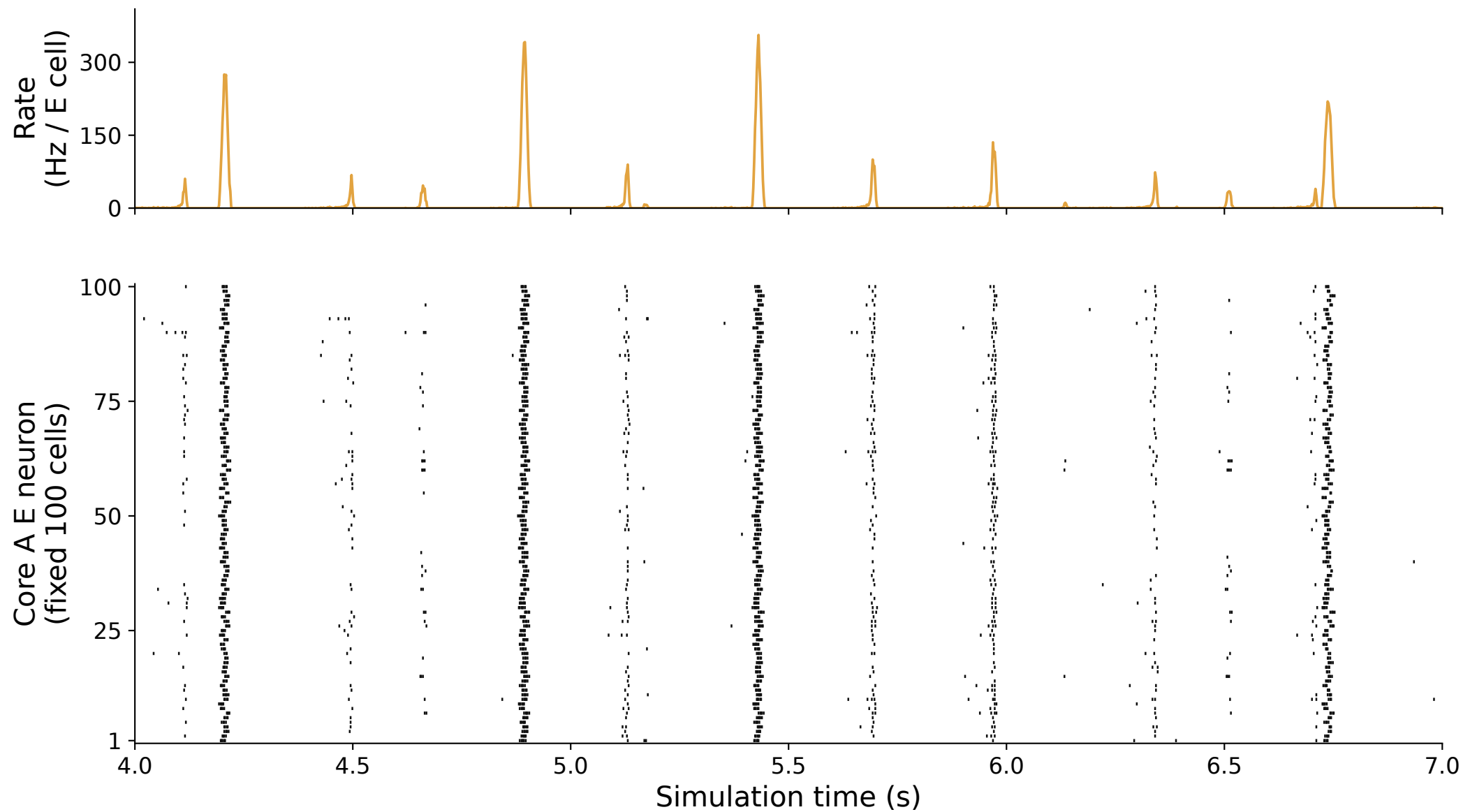
Irregular bursts | $E \rightarrow E = 0.7$

Rate: all 720 Core A E cells · raster: occupied 2 ms bins · same time and cells across states



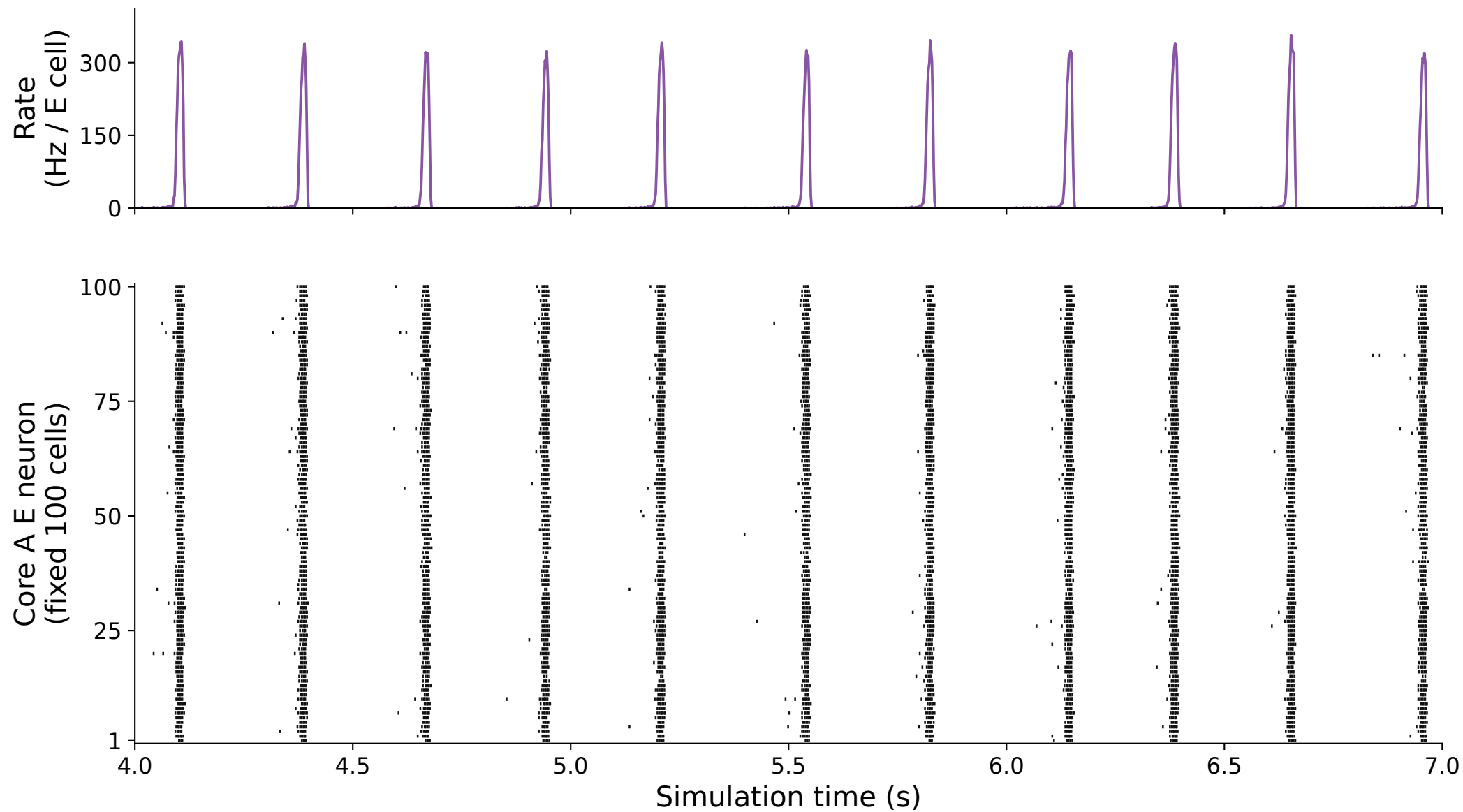
Intermediate | $E \rightarrow E = 0.85$

Rate: all 720 Core A E cells · raster: occupied 2 ms bins · same time and cells across states



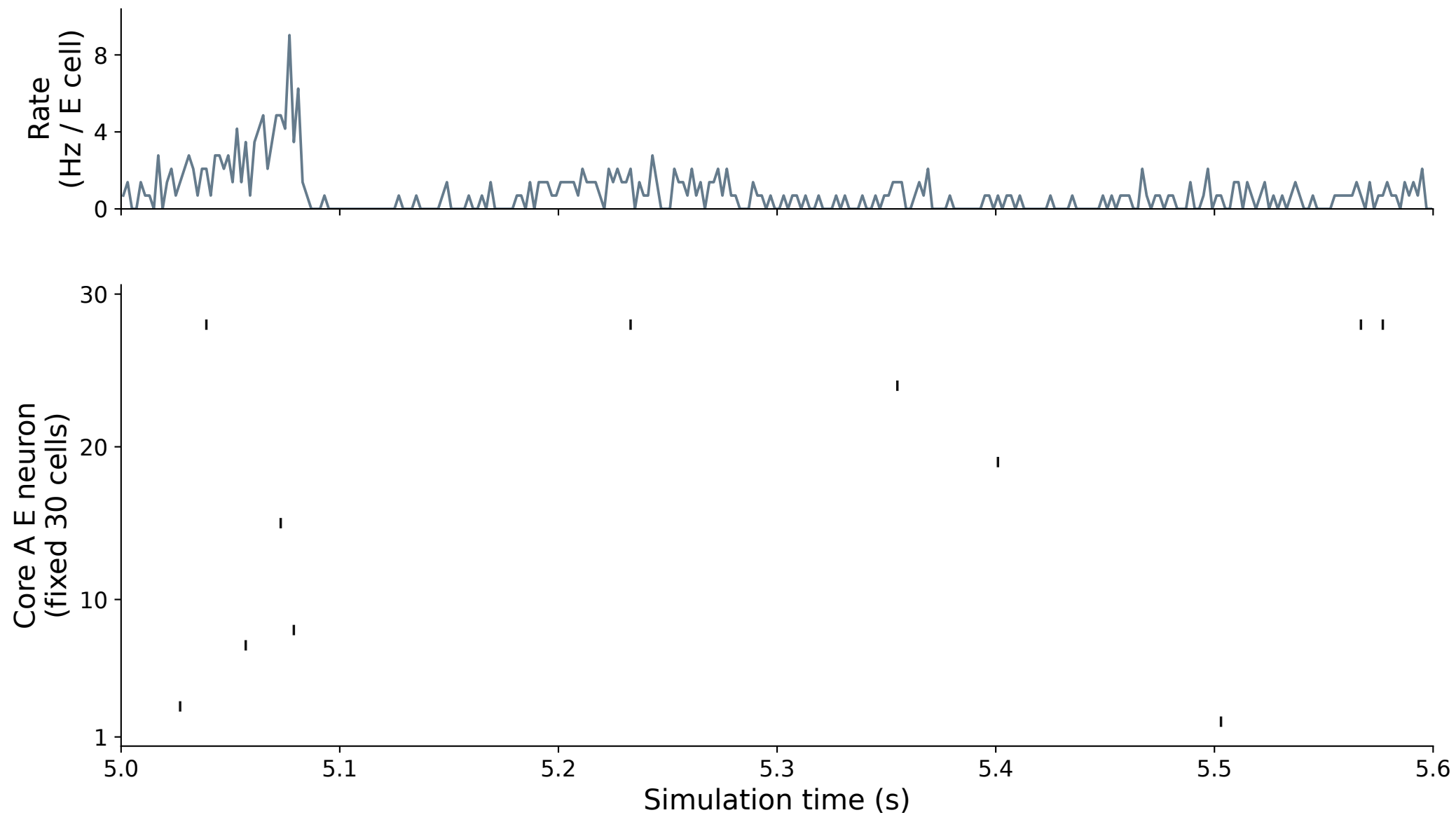
Regular bursts | $E \rightarrow E = 1$

Rate: all 720 Core A E cells · raster: occupied 2 ms bins · same time and cells across states



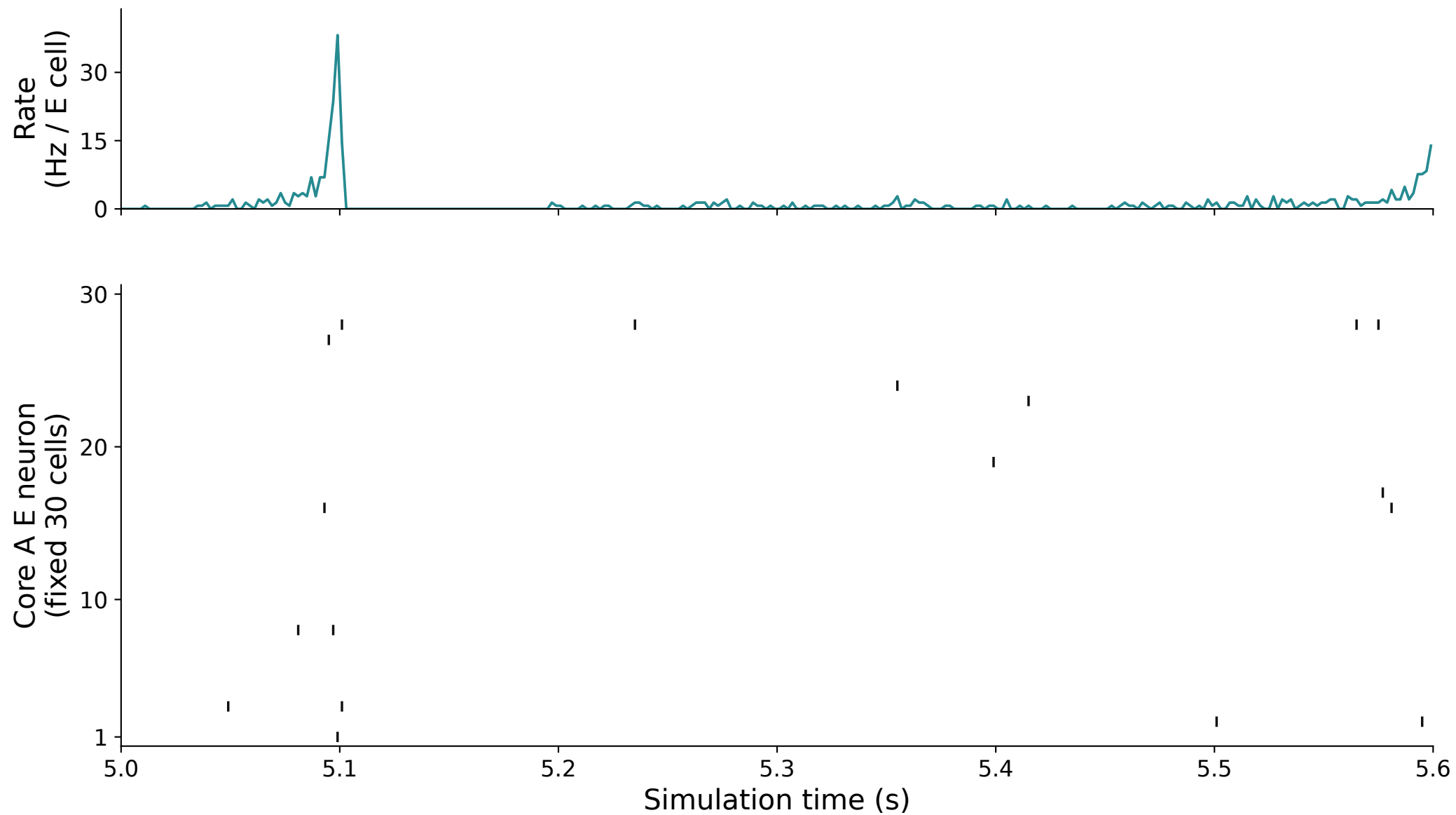
Low activity | $E \rightarrow E = 0.5$

Rate: all 720 Core A E cells · raster: occupied 2 ms bins · same time and cells across states



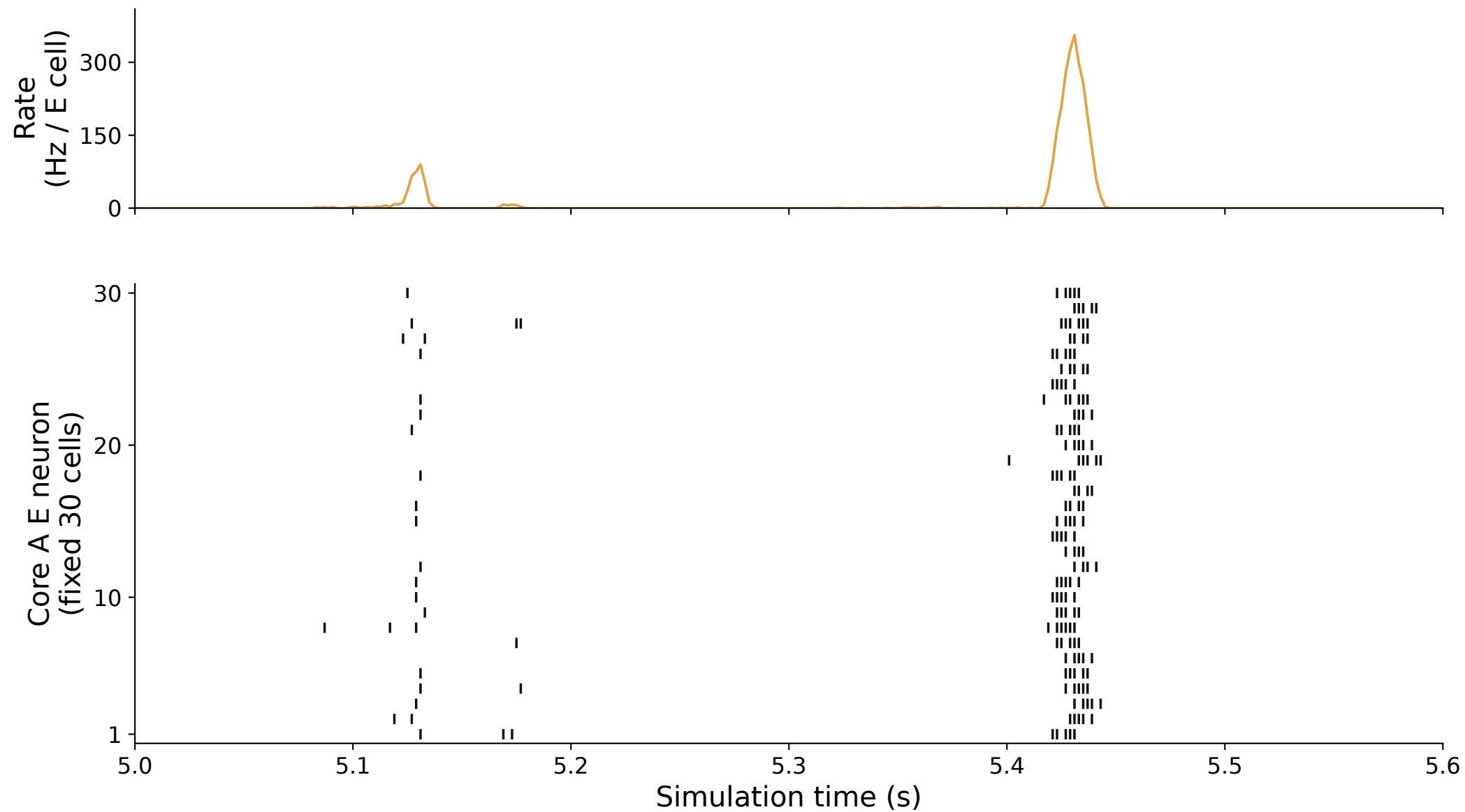
Irregular bursts | $E \rightarrow E = 0.7$

Rate: all 720 Core A E cells · raster: occupied 2 ms bins · same time and cells across states



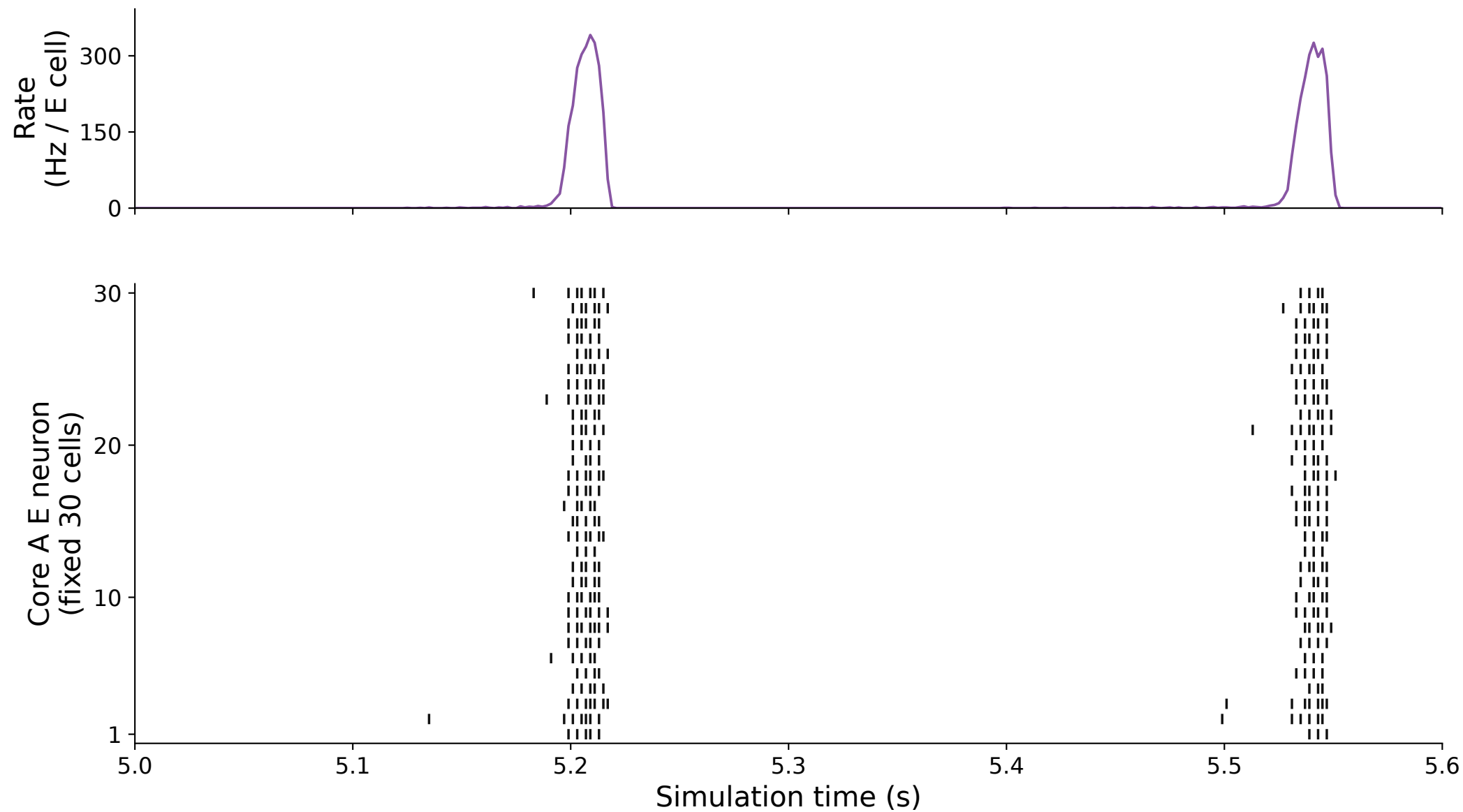
Intermediate | $E \rightarrow E = 0.85$

Rate: all 720 Core A E cells · raster: occupied 2 ms bins · same time and cells across states



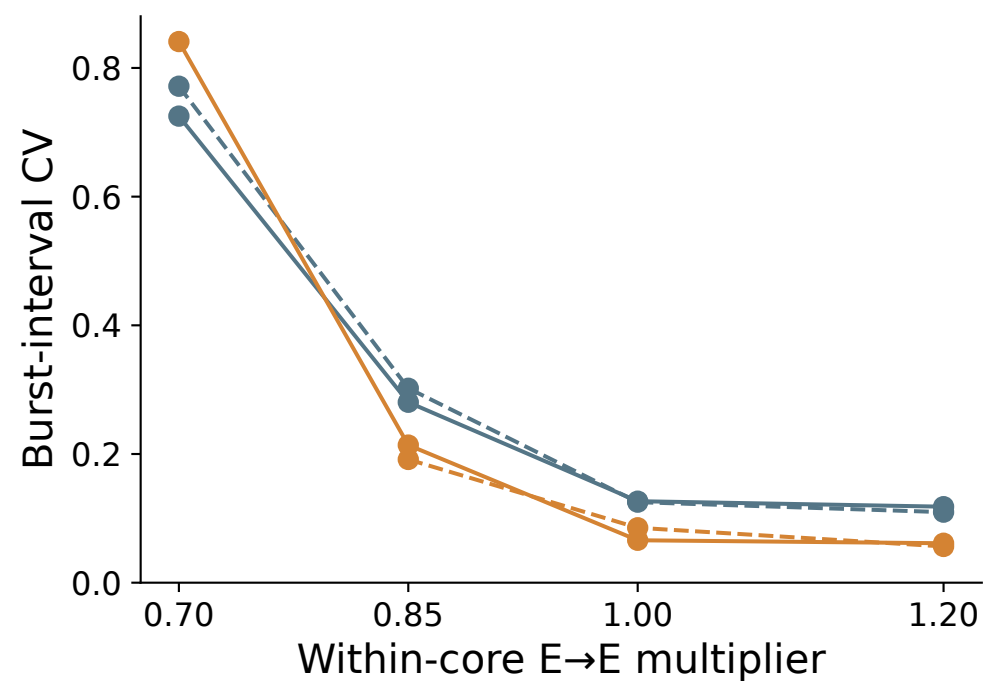
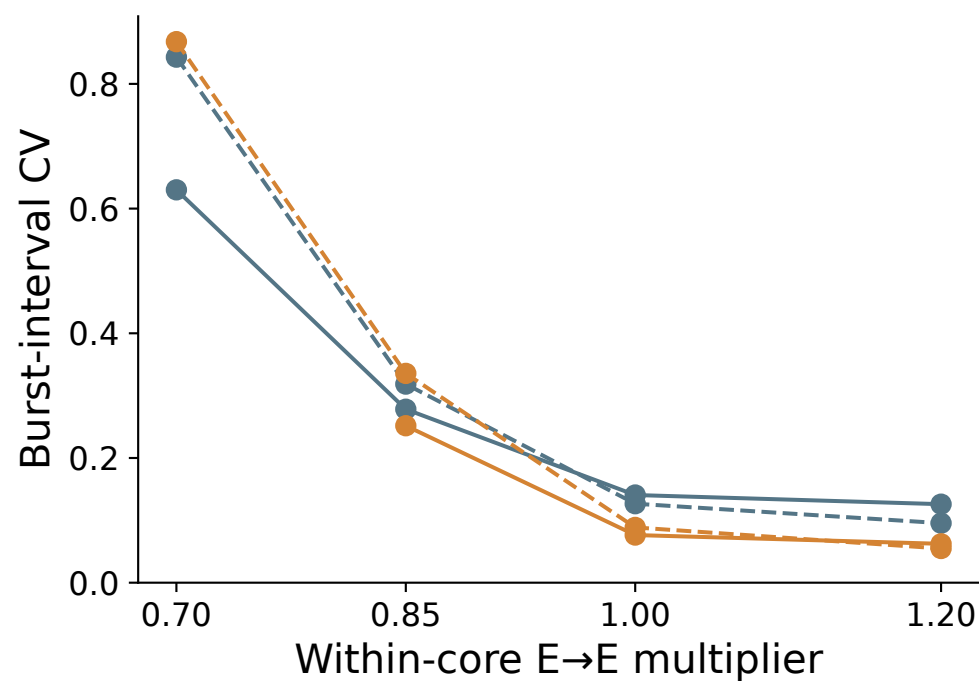
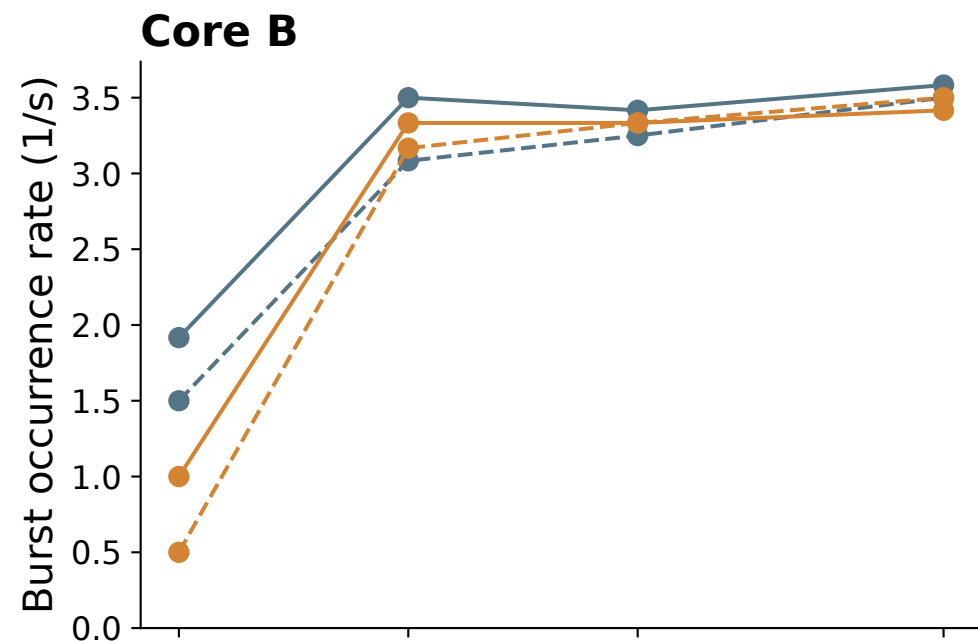
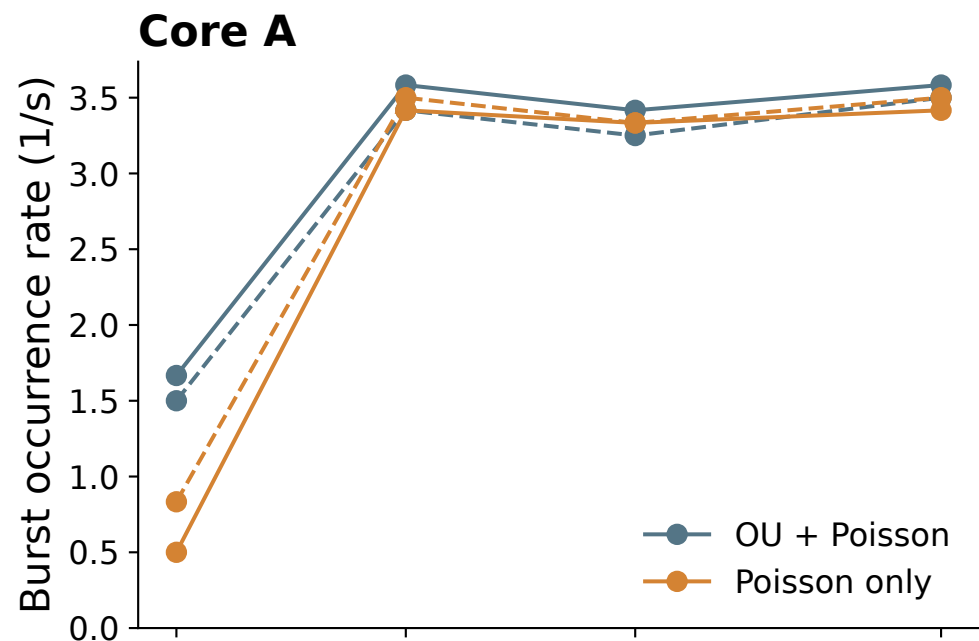
Regular bursts | $E \rightarrow E = 1$

Rate: all 720 Core A E cells · raster: occupied 2 ms bins · same time and cells across states



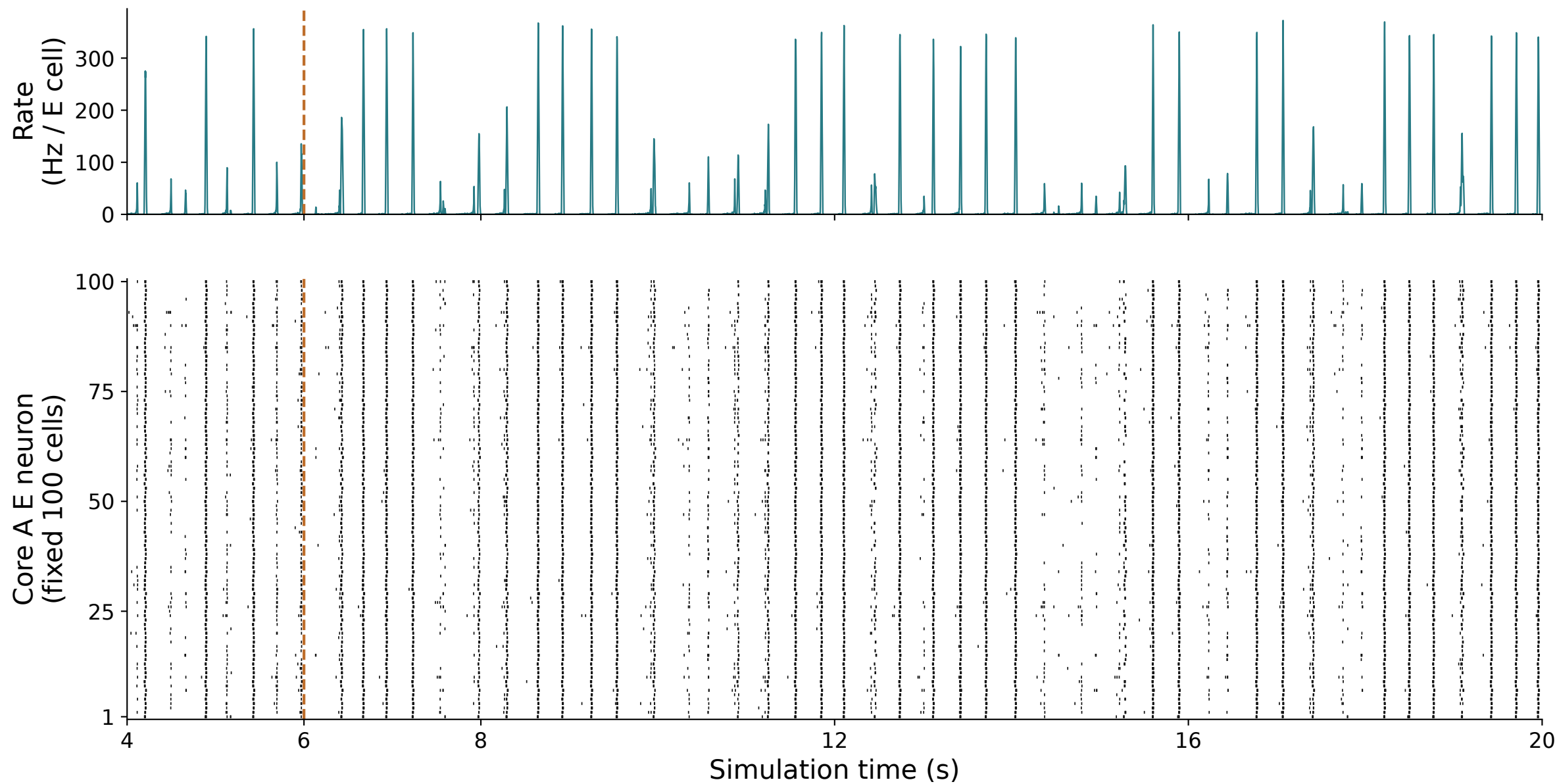
Does rhythmic bursting require slow core-wide OU input?

OU removed at 6 s; private Poisson retained · matched comparison window: 8–20 s



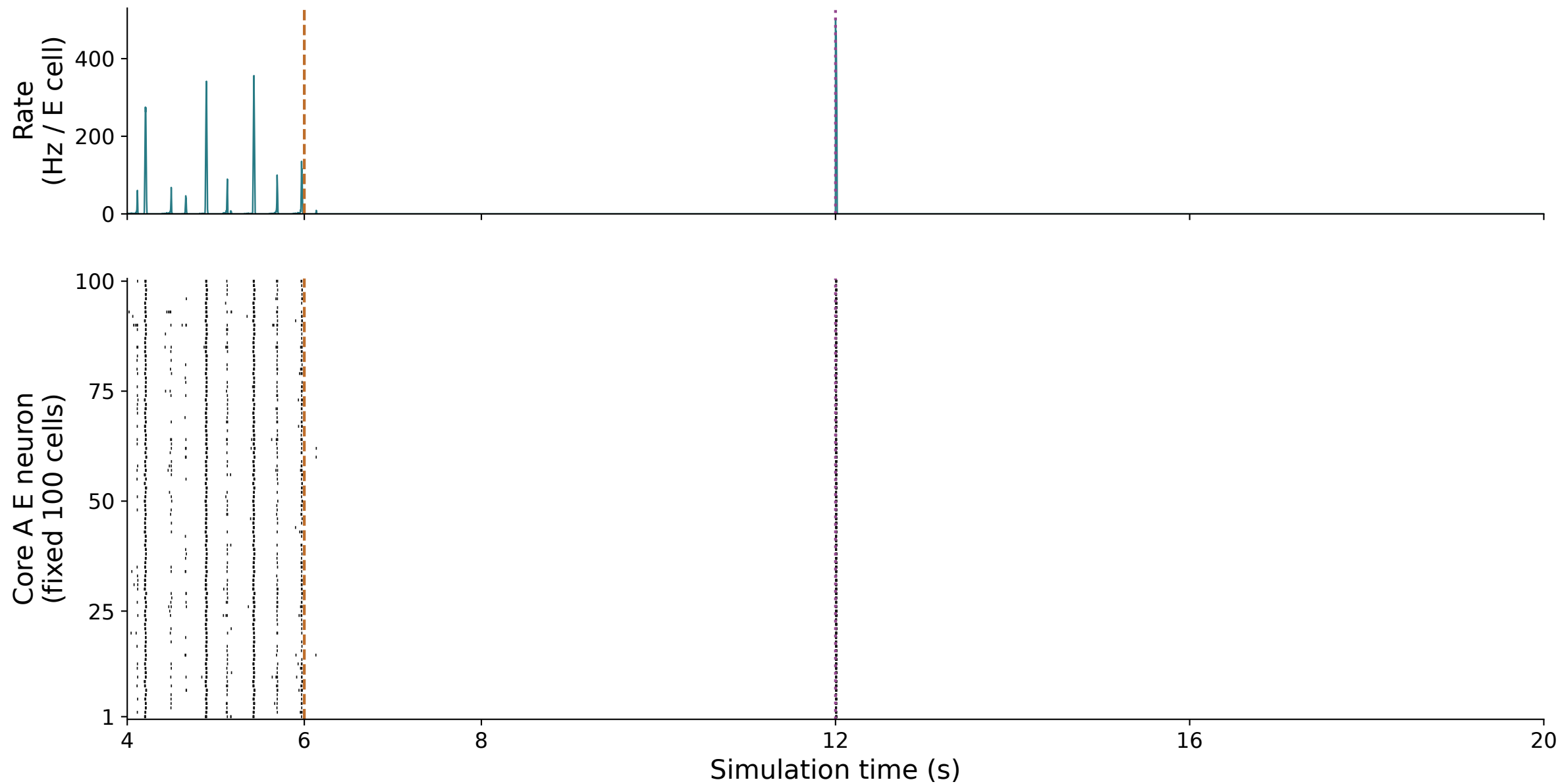
$E \rightarrow E = 0.85$ | OU removed; private Poisson retained

Continuous native state · actual 0.1 ms spike times · no Z/M dynamics



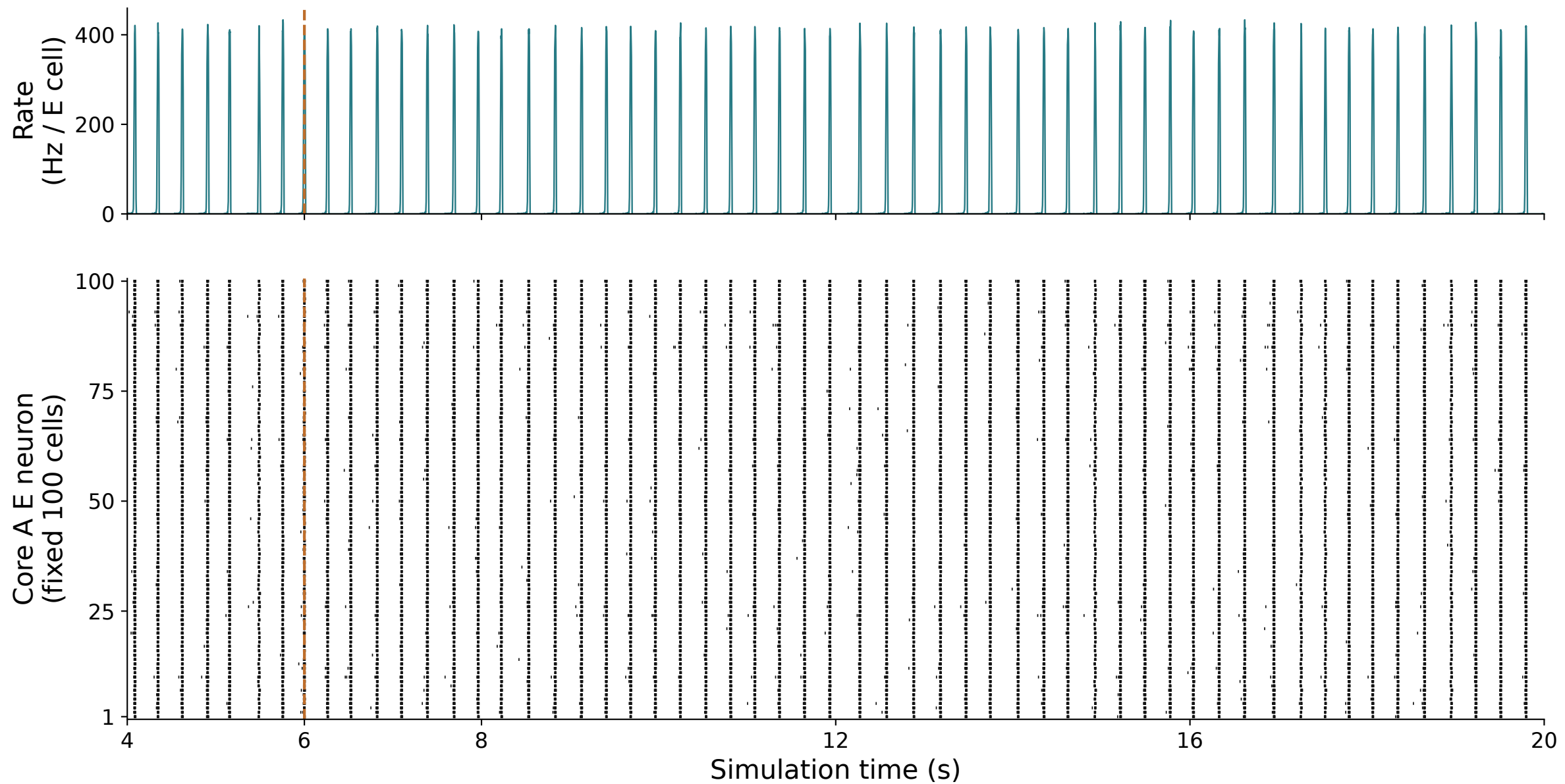
$E \rightarrow E = 0.85$ | All input fluctuations removed; finite pulse at 12 s

Continuous native state · actual 0.1 ms spike times · no Z/M dynamics



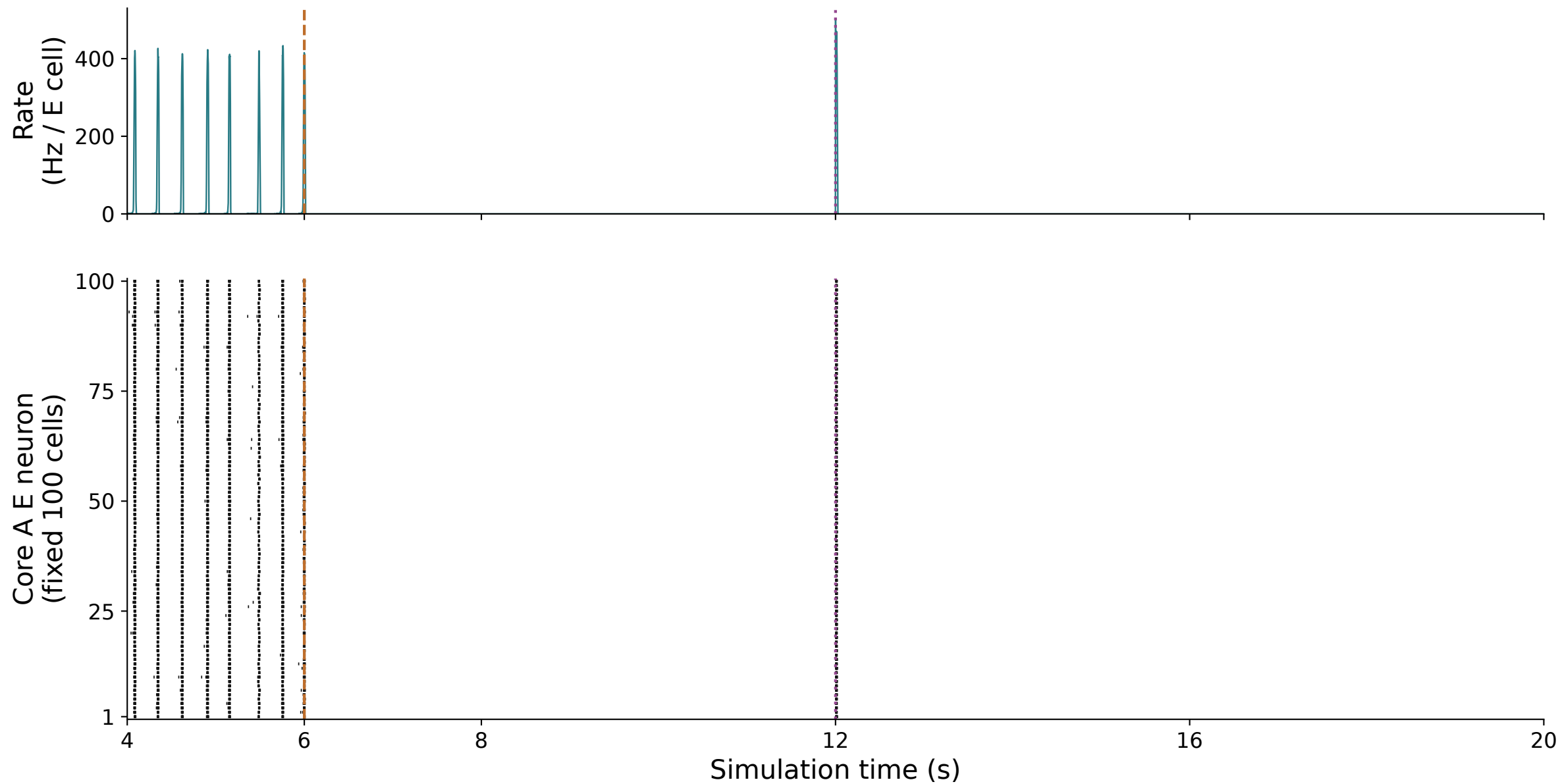
$E \rightarrow E = 1.2$ | OU removed; private Poisson retained

Continuous native state · actual 0.1 ms spike times · no Z/M dynamics

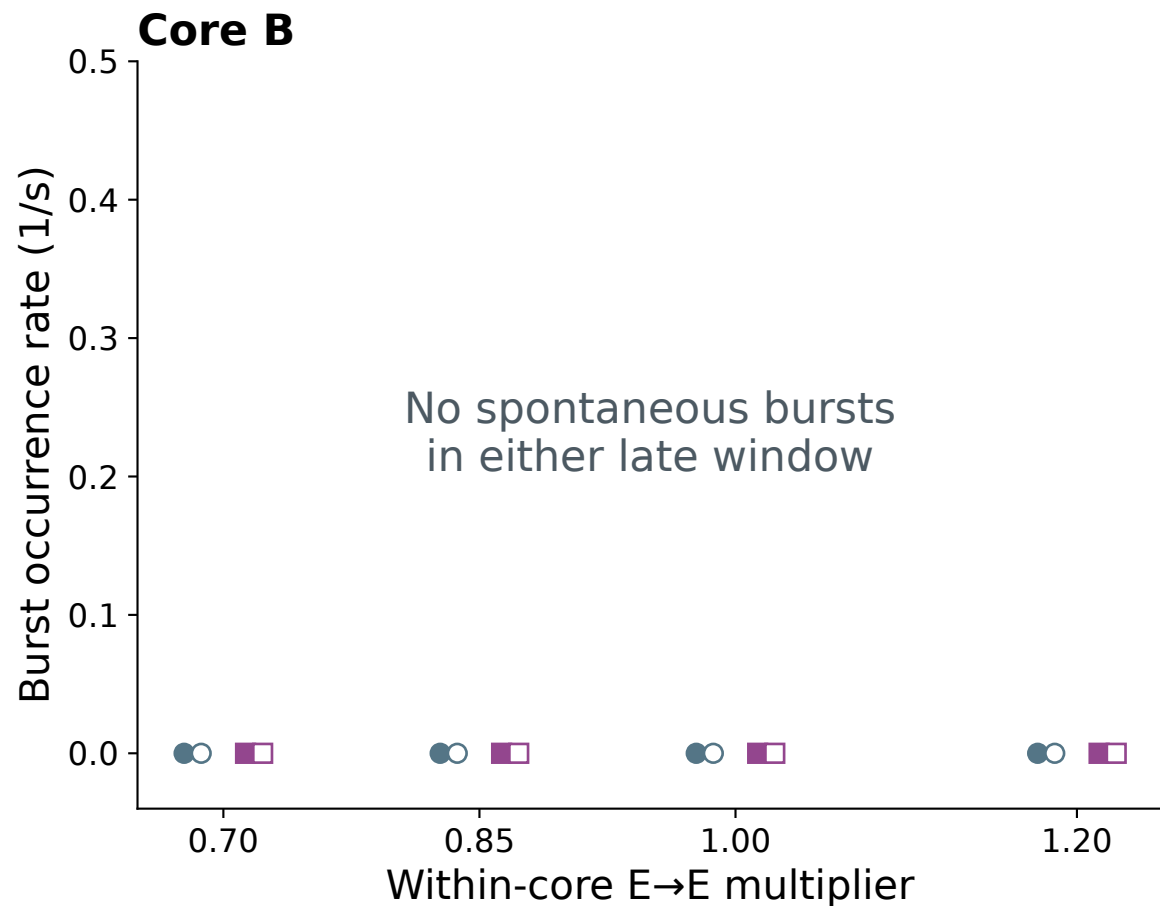
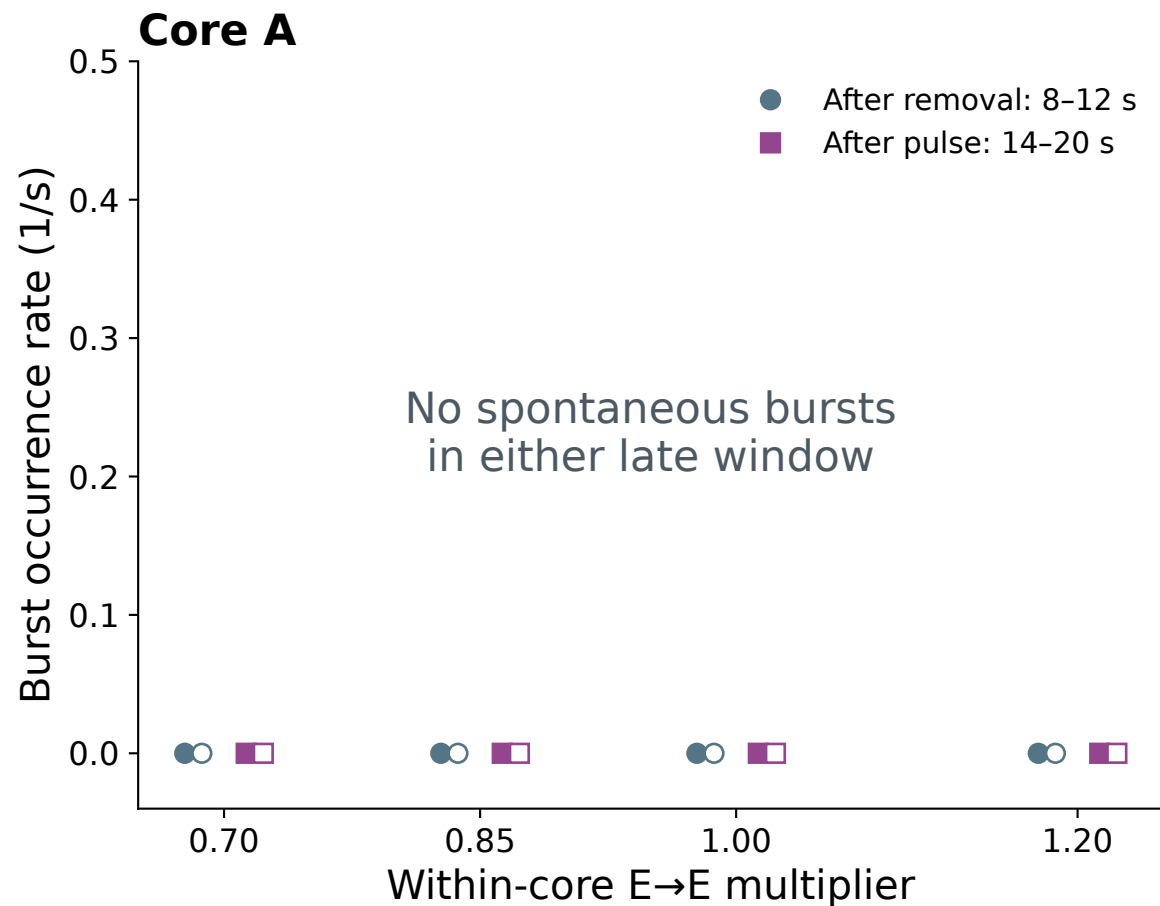


$E \rightarrow E = 1.2$ | All input fluctuations removed; finite pulse at 12 s

Continuous native state · actual 0.1 ms spike times · no Z/M dynamics

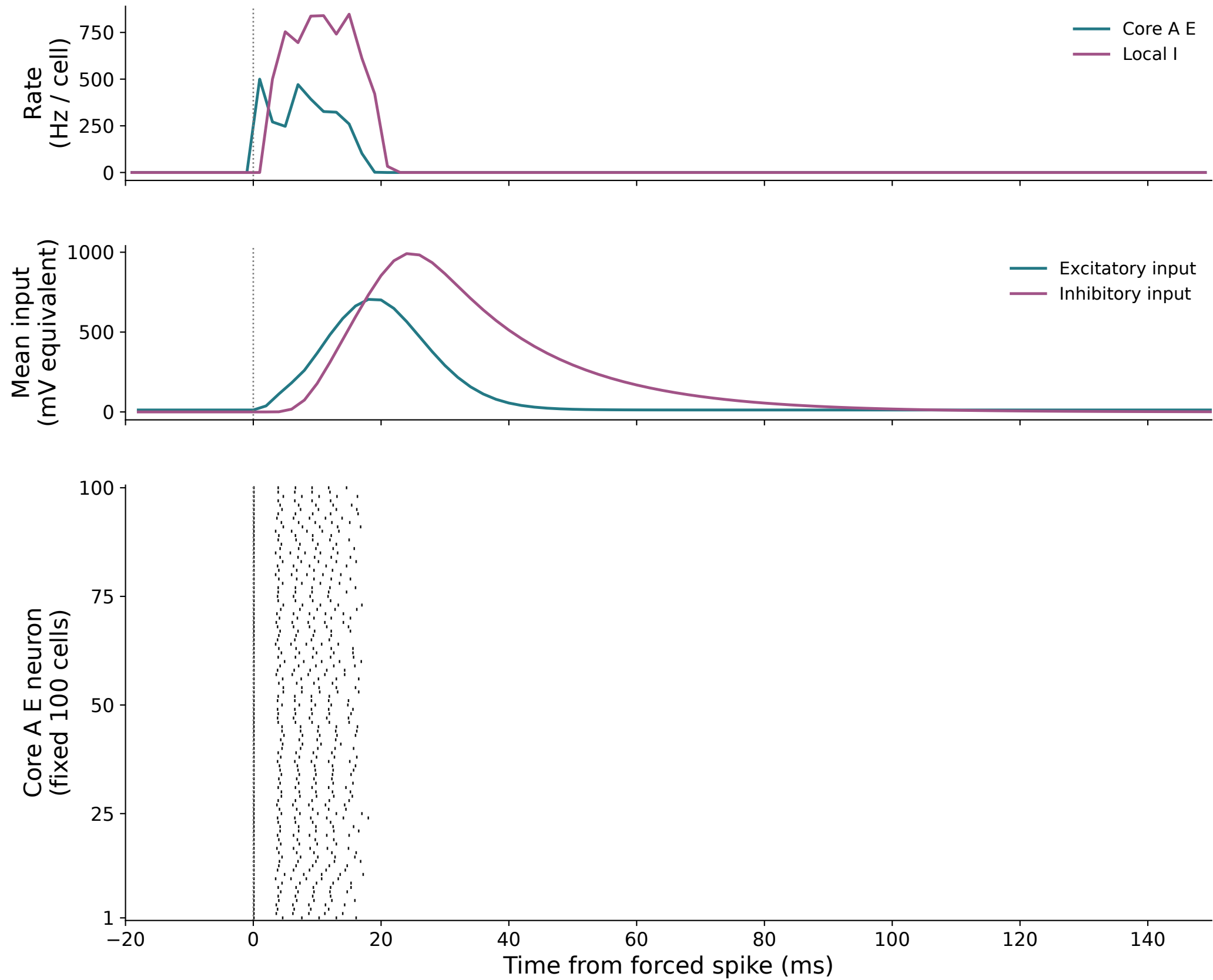


Finite-time persistence under deterministic input



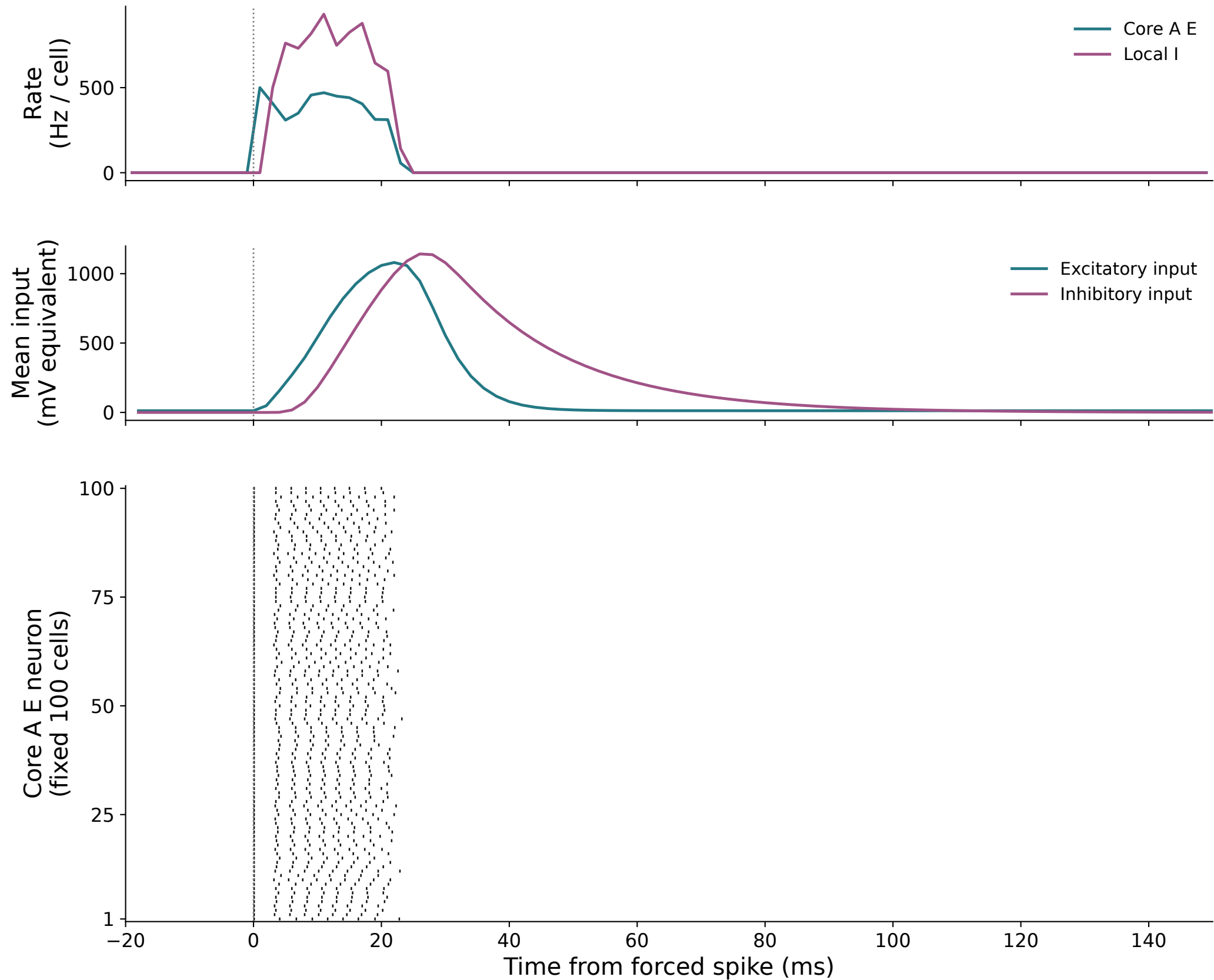
Finite pulse evokes a transient burst | $E \rightarrow E = 0.85$

One forced spike per Core A E cell at 12 s · constant external input thereafter



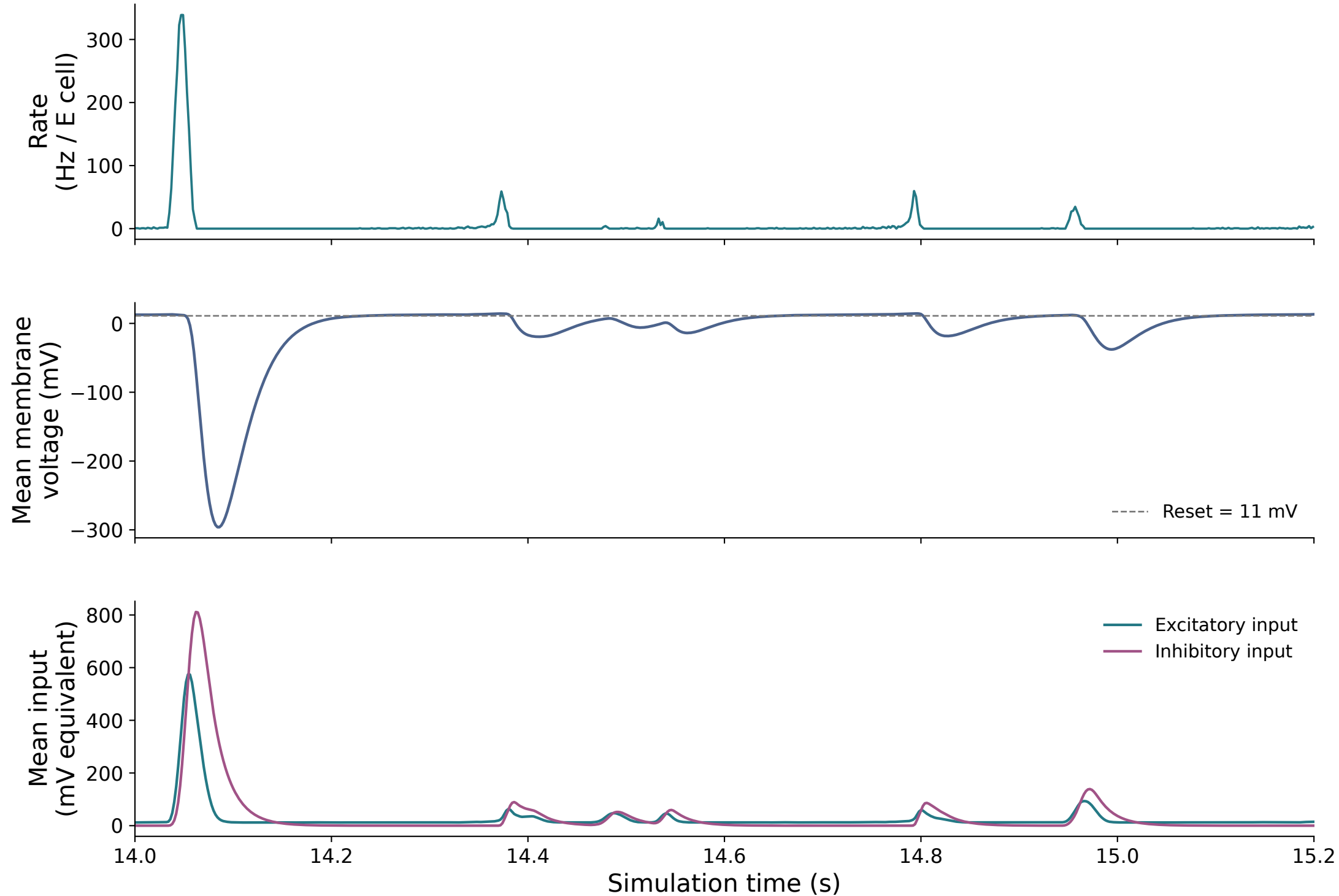
Finite pulse evokes a transient burst | $E \rightarrow E = 1.2$

One forced spike per Core A E cell at 12 s · constant external input thereafter



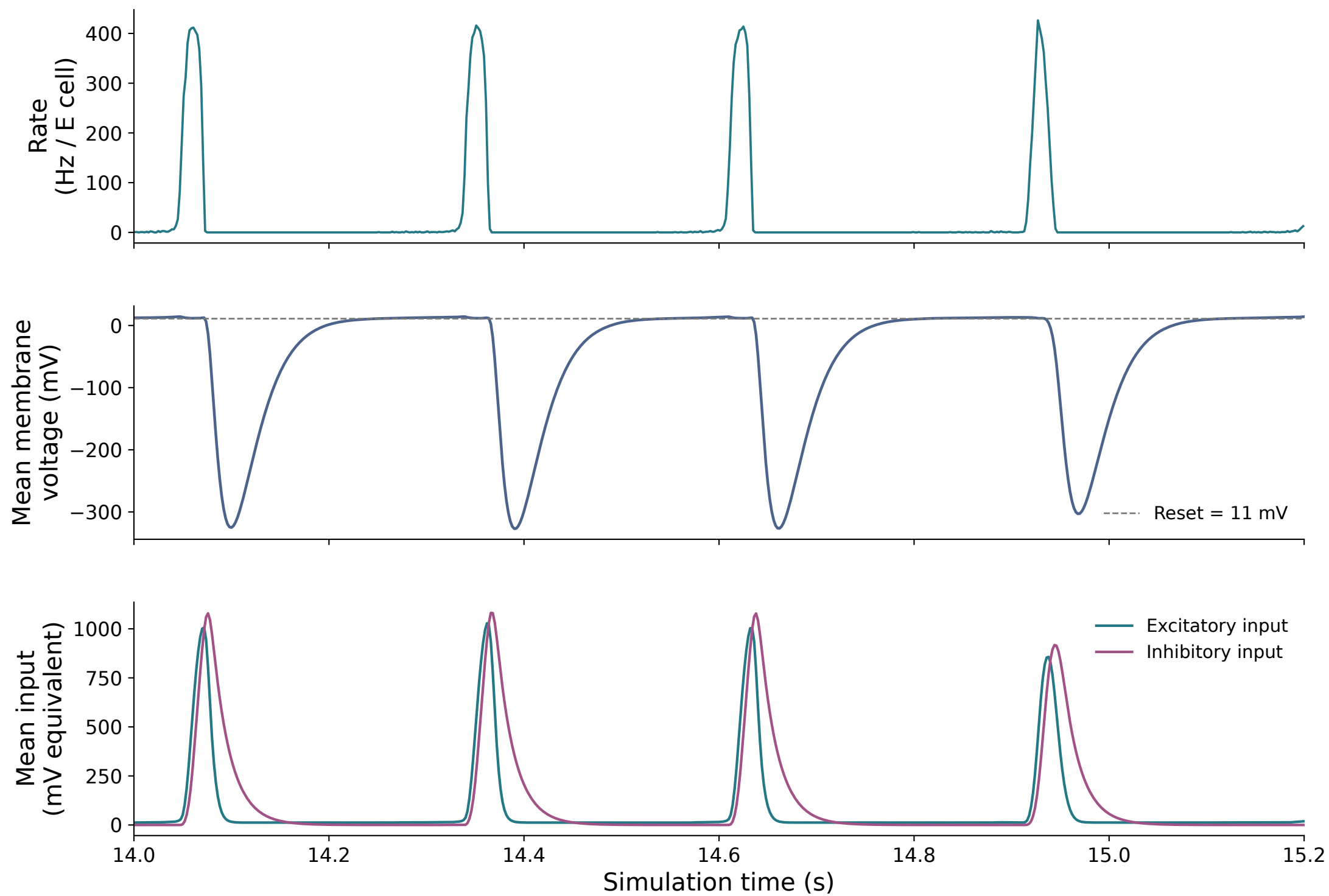
Native burst and recovery trajectory | $E \rightarrow E = 0.85$

Core A, Poisson only · current-based LIF · model voltage reference



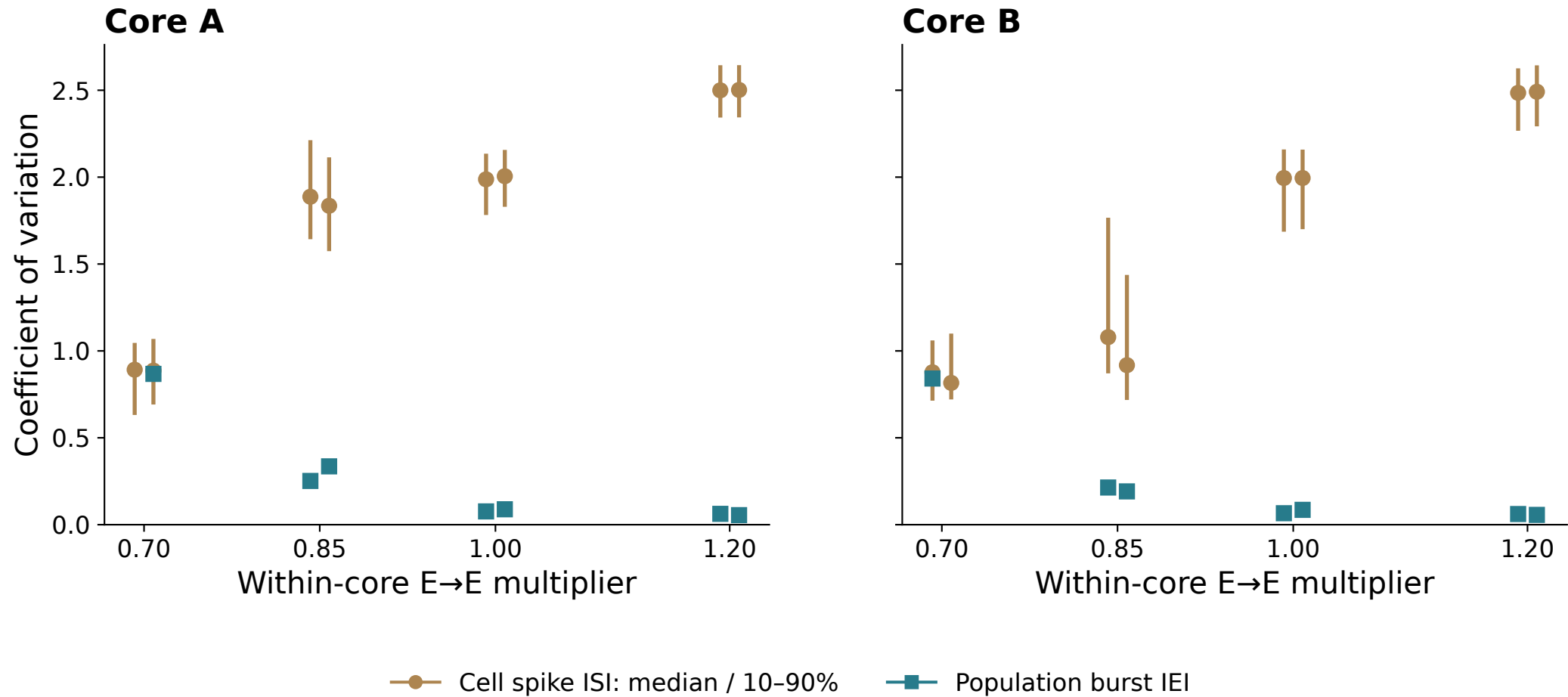
Native burst and recovery trajectory | $E \rightarrow E = 1.2$

Core A, Poisson only · current-based LIF · model voltage reference

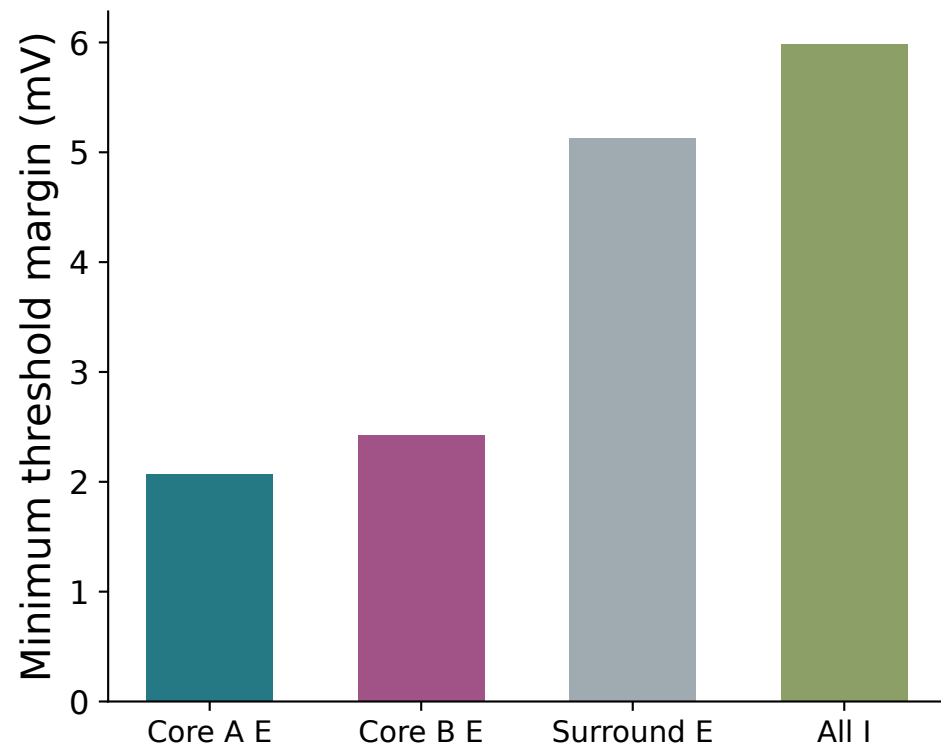
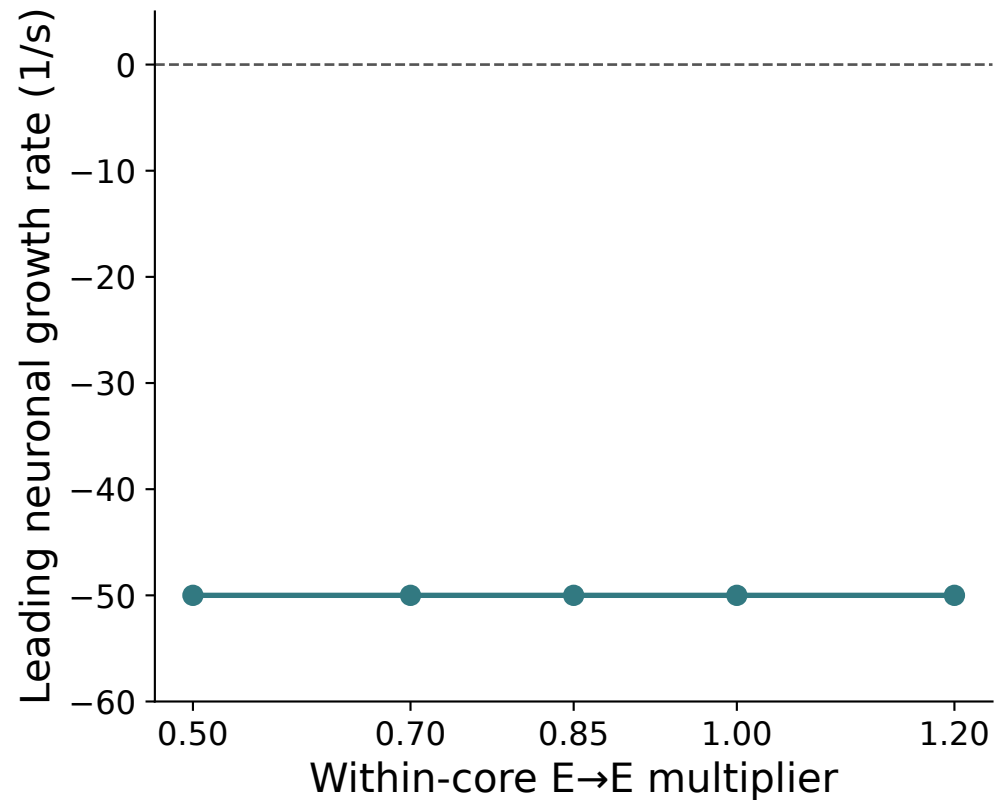


Cell irregularity and burst timing are different observables

Poisson only, 8–20 s · cell CV requires ≥ 21 spikes · cell bars describe eligible recorded cells

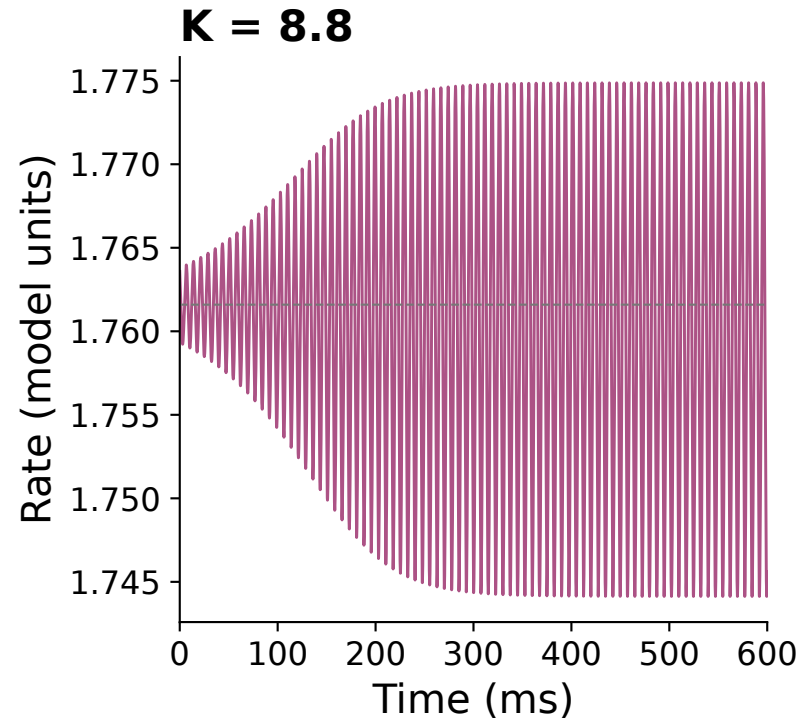
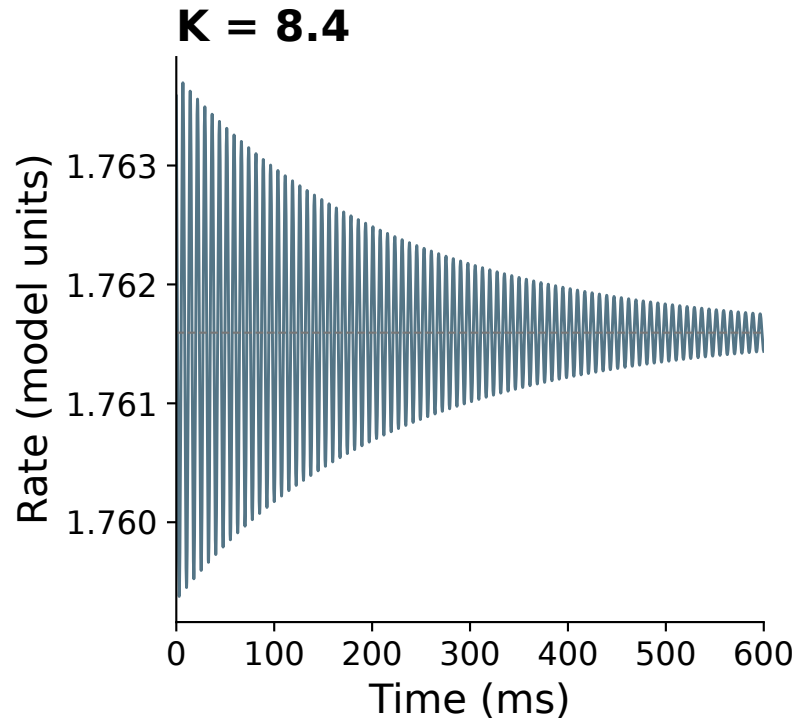
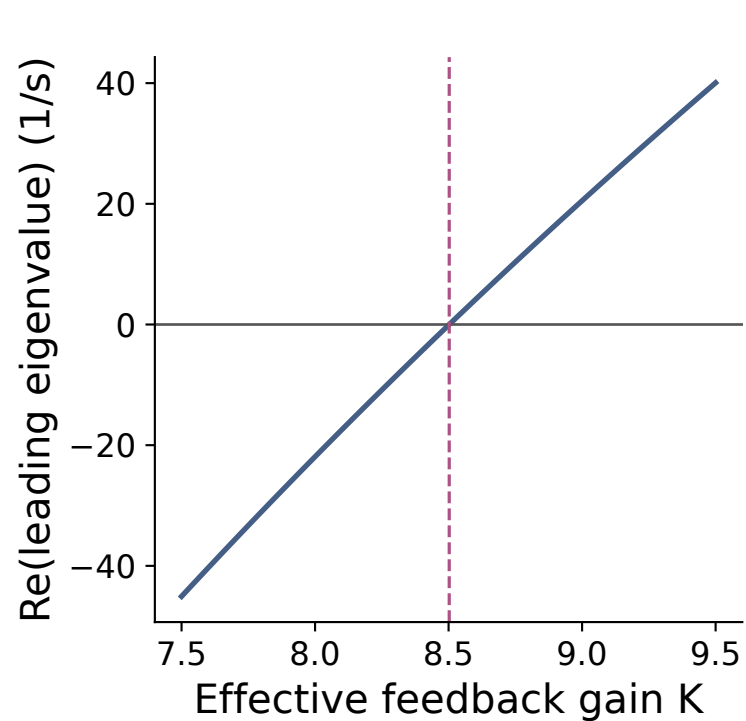


Exact silent branch of the native deterministic network



Positive threshold margin: no local spike feedback · this is not the noisy active population state

Literature benchmark: Brunel-Hakim 2008 delay-rate model



$K_c = 8.5024$ · onset frequency = 134.38 Hz · not a fitted reduction of our SNN